Extremal Combinatorics Notes

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1 Turán numbers, Mantel, and crude bounds (Jan 12)

1.1 Basic Definitions

Definition 1.1. *Turan Numbers (Forbidden Subgraph Problems): F is a graph and G graph is F-free if G contains no copy of F as a subgraph.*

We want to maximize the size of G subject to G being F-free. Where size=$e(G)$ = # edges in G

Definition 1.2. $ex(n, F) = \max\{e(G) | G$ *is F-free and G is n-vertex graph*$\}$

1.2 Crude Turán's Theorem

Theorem 1.3. *Crude Turan's Theorem: If G is n-vertex graph,*

$$e(G) = \epsilon\binom{n}{2}.$$

then G contains an independent set of size greater than or equal $\frac{1}{2}\frac{1}{\epsilon} + \frac{1}{2}$
In particular, if $e(G) \geq (1-\epsilon)\binom{n}{2}$ edges, then G contains a clique on greater than or equal $\frac{1}{2\epsilon} + \frac{1}{2}$ vertices.

$$ex(n, K_t) \leq \left(1 - \frac{1}{2t-1}\right)\binom{n}{2}.$$

Proof. Pick k vertices in G at random without replacement. Say we pick $v_1, ..., v_k$
For each edge between v_i and v_j, delete v_i.
Let $X = \#$ deleted vertices
Note: $X \leq e(G[v_1, ..., v_k]) =: Y$

$$\mathbb{E}[Y] = \sum_{1 \leq i \leq j \leq k} \mathbb{P}[v_i v_j \in e(G)] = \binom{k}{2}\frac{\epsilon\binom{n}{2}}{\binom{n}{2}}s = \epsilon\binom{k}{2}.$$

Thus, there is a choice of $v_1, ..., v_k$ such that $Y \leq \epsilon\binom{k}{2}$. Fix it. Let $\alpha(G)$ = size of largest independent set in G satisfies $\alpha(G) \geq k - Y \geq k - \epsilon\binom{k}{2}$

$$\left(k - \epsilon\binom{k}{2}\right)' = 1 + \frac{\epsilon}{2} - \epsilon k.$$

Set $k = \frac{1+\frac{\epsilon}{2}}{2} + \delta$ for some $|\delta| \leq \frac{1}{2}$.

$$\begin{aligned}
\alpha(G) &\geq \frac{1}{\epsilon} + \frac{1}{2} + \delta - \epsilon\left(\left(\frac{1}{\epsilon} + \frac{1}{2} + \delta\right)\left(\frac{1}{\epsilon} - \frac{1}{2} + \delta\right)\right)/2 \\
&= \frac{1}{\epsilon} + \frac{1}{2} + \delta - \epsilon\frac{(\frac{1}{\epsilon} + \delta)^2 - \frac{1}{4}}{2} \\
&= \frac{1}{\epsilon} + \frac{1}{2} + \delta - \epsilon\frac{(1/\epsilon)^2 + 2\frac{1}{\epsilon}\delta + \delta^2 - \frac{1}{4}}{2} \\
&= \frac{1}{2}\frac{1}{\epsilon} + \frac{1}{2} + \frac{\frac{1}{4} - \delta^2}{2} \geq \frac{1}{2}\frac{1}{\epsilon} + \frac{1}{2}
\end{aligned}$$

□

1.3 Mantel's and Turán's Theorems

Remark 1.4. *Recall: k-uniform hypergraph on vertex set V is a subset of*

$$\binom{V}{k} := \{e \subseteq V : |e| = k\}.$$

Example 1.5. *In k-uniform hypergraph on vertex with $\epsilon\binom{n}{k}$ edges there is an independent set of size $\geq c_k \frac{1}{\epsilon}^{\frac{1}{k-1}}$*

Theorem 1.6. *Mantel's Theorem: $ex(n, K_3) = \lfloor \frac{n^2}{4} \rfloor$*
More generally, define Turan's graph $T_{n,r}$ to be the n-vertex complete r-partite graph with part sizes as equal as possible.
Note: $K_{r+1} \not\subseteq T_{n,r}$ because among $r+1$ vertices some pair is in same part.

Theorem 1.7. *Turan's Theorem: $ex(n, K_{r+1}) = e(T_{n,r})$ and furthermore $T_{n,r}$ is the only extremizer.*

$$e(T_{n,r}) = \left(1 - \frac{1}{r}\right)\binom{n}{2} + O(n).$$

Proof. Let G be a graph on n vertices that is K_{r+1}-free and has the maximum possible number of edges (an extremizer). We proceed in two steps.

Suppose G is not a complete multipartite graph. Then non-adjacency is not an equivalence relation, implying there exist two non-adjacent vertices x, y that do not have identical neighborhoods.

Assume $\deg(x) \leq \deg(y)$. Delete x and then clone y to obtain y'. The new graph G' has

$$e(G') = e(G) - \deg(x) + \deg(y) \geq e(G).$$

Also, G' is K_{r+1}-free because any copy of K_{r+1} has at most one of y or y' (since they are non-adjacent). If a clique contains y', it can be swapped for y (since $N(y') \subseteq N(y)$), which would imply $K_{r+1} \subseteq G$, a contradiction.

Repeating this process transforms G into a complete multipartite graph without decreasing edges. Thus, the extremizer must be a complete multipartite graph.
Since G is K_{r+1}-free, it must be k-partite with $k \leq r$. To maximize edges, we set $k = r$. Let the parts have sizes $n_1, \ldots, n_r$.

$$e(G) = \binom{n}{2} - \sum_{i=1}^{r} \binom{n_i}{2}$$

To maximize $e(G)$, we must minimize the subtracted sum of pairs inside the parts. This occurs when n_i are as equal as possible ($|n_i - n_j| \leq 1$). If $n_i \geq n_j + 2$, moving a vertex from part i to part j increases the edge count by $(n_i - 1) - n_j > 0$.

Thus, the unique extremizer is the balanced complete r-partite graph, $T_{n,r}$. $\square$

2 Turán density, supersaturation, and induction (Jan 14)

2.1 Inductive Proof of Turán's Theorem

Proof. Second proof of Turan's Theorem with induction on r.
(Induction Step) Let x be a vertex of largest degree, let $A = N(x)$ be its neighborhood.

$$B := V(G) \setminus A.$$

We have $e(G) = e(A) + e(A, B) + e(B)$ and A is K_{r-1}-free so $e(A) \leq e(T_{|A|,r})$
Each vertex in B has deg $\leq \deg(x) = |A|$
So,

$$e(A, B) + e(B) \leq \sum_{v \in B} \deg(v) \leq |B||A|.$$

So,

$$e(G) \leq e(T_{|A|,r}) + |A||B| \leq e(T_{n,r}).$$

The right hand side is equal to number of edges in complete r-partite graph with one part of size $|B|$ and other being as equal as possible.
Equality holds iff B is independent and A must be a copy of $e(T_{|A|,r-1})$. Hence G is $T_{n,r}$ for equality to hold. □

2.2 Monotonicity of Extremal Density

Theorem 2.1. *Observe that if F is a graph then*

$$\frac{ex(n, F)}{\binom{n}{2}} \geq \frac{ex(n+1, F)}{\binom{n+1}{2}}.$$

Proof. Pick n vertices in an F-free $(n+1)$ vertex graph G at random without replacement; let G' be the induced graph on these n vertices.

$$e(G') \leq ex(n, F).$$

$$\mathbb{E}(e(G')) = \binom{n}{2} \frac{e(G)}{\binom{n+1}{2}} \implies \frac{e(G)}{\binom{n+1}{2}} \leq \frac{ex(n, F)}{\binom{n}{2}}.$$

□

Definition 2.2.

$$\pi(F) := \lim_{n \to \infty} \frac{ex(n, F)}{\binom{n}{2}}.$$

The Turan density of F is

$$\pi(K_{r+1}) = 1 - \frac{1}{r}.$$

2.3 Counting Copies Above Threshold

Theorem 2.3. *Let F be any graph and let $\epsilon > 0$. Let G be an n-vertex graph. There is a $\delta > 0$ such that*

$$e(G) \geq (\pi(F) + \epsilon) \binom{n}{2} \text{ and } n \geq n_0(F, \epsilon).$$

then there are at least $\delta n^{|V(F)|}$ copies of F.

Proof. Let n_0 be large enough such that $\frac{ex(n_0,F)}{\binom{n_0}{2}} \leq \pi(F) + \frac{\epsilon}{2}$. Given n-vertex G as in theorem. Pick n_0 vertices at random without replacement. Let G' be the n_0-vertex graph induced on these chosen vertices.

$$\mathbb{E}[e(G')] = \frac{e(G)}{\binom{n}{2}} \binom{n_0}{2} \geq (\pi(F) + \epsilon) \binom{n_0}{2}.$$

Say G' is good if

$$e(G') \geq \left(\pi(F) + \frac{\epsilon}{2}\right)\binom{n_0}{2}.$$

$$\begin{aligned}\mathbb{E}(e(G')) &\leq \mathbb{P}(G' \text{ is bad})\left(\pi(F) + \frac{\epsilon}{2}\right)\binom{n_0}{2} + \mathbb{P}(G' \text{ is good})\binom{n_0}{2} \\ &\leq \left(\pi(F) + \frac{\epsilon}{2}\right)\binom{n_0}{2} + \mathbb{P}(G' \text{ is good})\binom{n_0}{2}\end{aligned}$$

Then $\frac{\epsilon}{2}\binom{n_0}{2} \leq \mathbb{P}(G' \text{ is good})\binom{n_0}{2}$ so $P(G' \text{ is good}) \geq \frac{\epsilon}{2}$
Pick $|V(F)|$ many vertices without replacement inside G', call the subgraph they induce G''. So

$$G \supset G' \supset G''.$$

$$\mathbb{P}(G'' \text{ contains } F) \geq \mathbb{P}(G' \text{ is good}) \cdot \mathbb{P}(G'' \text{ contains } F | G' \text{ is good}) \geq \frac{\epsilon}{2}\frac{1}{\binom{n_0}{|V(F)|}}.$$

Note that G'' is uniformly randomly chosen $|V(F)|$-vertex subset of G. This implies the # copies of F in $G \geq \mathbb{P}(G''\text{contains F})\binom{n}{|V(F)|} \geq P(...)\frac{n^{|V(F)|}(1-o(1))}{|V(F)|!}$. We can choose $\delta = \frac{1}{\binom{n_0}{|V(F)|}}$ $\square$

3 Erdős–Stone, Kővári–Sós–Turán, and geometry (Jan 16)

3.1 Erdős–Stone–Simonovits Theorem

Theorem 3.1 (Erdős-Stone-Simonovits)**.** *Let $T_{n,r}$ denote the Turán graph. We know that:*

$$ex(n, T_{s,(r+1)}) = \left(1 - \frac{1}{r} + o(1)\right)\binom{n}{2}.$$

This implies that if H is any graph with chromatic number $\chi(H) = r + 1$, then

$$ex(n, H) \leq \left(1 - \frac{1}{r} + o(1)\right)\binom{n}{2}.$$

Conversely, if $\chi(H) > r$ (i.e., H is not r-partite), then $T_{n,r}$ is H-free. Hence,

$$ex(n, H) \geq e(T_{n,r}) = \left(1 - \frac{1}{r} + o(1)\right)\binom{n}{2}.$$

Combining these, for any non-bipartite graph H:

$$ex(n, H) = \left(1 - \frac{1}{\chi(H) - 1} + o(1)\right)\binom{n}{2}.$$

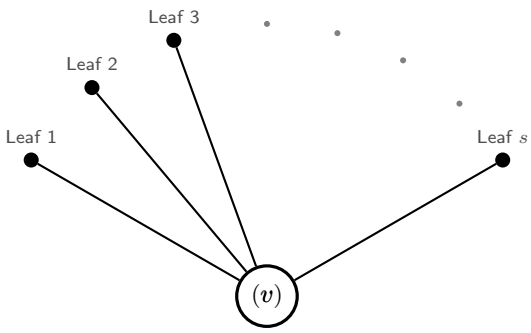
It is important to note that this theorem holds for bipartite graphs as well, where $\chi(H) = 2$. However, substituting $r = \chi(H) - 1 = 1$ gives a coefficient of $1 - 1/1 = 0$. This implies that $ex(n, H) = o(n^2)$, but it does not provide the precise asymptotic behavior (e.g., $n^{2-\epsilon}$). Therefore, while valid, the Erdős-Stone-Simonovits theorem is not the primary tool for determining extremal numbers of bipartite graphs; we rely on theorems like Kővári-Sós-Turán for those bounds.

3.2 Kővári–Sós–Turán Bound

Theorem 3.2 (Kővári-Sós-Turán 1954)**.**

$$ex(n, K_{s,t}) \leq \frac{1}{2}(t-1)^{\frac{1}{s}} \cdot n^{2-\frac{1}{s}} + O(n).$$

Proof. Let G be a $K_{s,t}$-free graph with n vertices. We will count X, the number of copies of $K_{1,s}$ (stars with s leaves).



We count this in two ways:

1. Since there is no $K_{s,t}$, any set of s vertices can have at most $t - 1$ common neighbors. Thus:

$$X \leq \binom{n}{s}(t-1).$$

2. Let d_v be the degree of vertex v. Then:

$$X = \sum_{v \in V(G)} \binom{d_v}{s}.$$

A simple-minded application of Jensen's inequality here fails because the binomial coefficient $\binom{x}{s}$ is not convex for all x. To fix this, we define the function:

$$f_s(x) = \begin{cases} \binom{x}{s} & \text{if } x \geq s-1 \\ 0 & \text{otherwise.} \end{cases}$$

Note that $f_s(x)$ is convex for all x because its derivative:

$$\frac{d}{dx} f_s(x) = \begin{cases} \left(\frac{1}{x} + \frac{1}{x-1} + \cdots + \frac{1}{x-s+1}\right)\binom{x}{s} & \text{if } x > s-1 \\ 0 & \text{otherwise} \end{cases}$$

is non-decreasing. Since d_v is an integer, $\binom{d_v}{s} = f_s(d_v)$. We can now validly apply Jensen's inequality:

$$X = \sum_{v \in V} f_s(d_v) \geq n f_s(d_{\text{avg}}) = n f_s\left(\frac{2m}{n}\right),$$

where $m = e(G)$.

Combining (1) and (2):

$$n f_s\left(\frac{2m}{n}\right) \leq \binom{n}{s}(t-1).$$

If $m \leq \frac{1}{2}(s-1)n$, the bound holds trivially. Otherwise, we approximate $\binom{x}{s} \approx \frac{x^s}{s!}$:

$$n\frac{\left(\frac{2m}{n} - s\right)^s}{s!} \leq (t-1)\frac{n^s}{s!}.$$

Rearranging yields:

$$\frac{2m}{n} \leq (t-1)^{\frac{1}{s}} n^{1-\frac{1}{s}} + s \implies m \leq \frac{1}{2}(t-1)^{\frac{1}{s}} n^{2-\frac{1}{s}} + O(n).$$

□

3.3 Unit Distance Problem

Example 3.3 (Unit Distance Problem)**.** *How can we place n points in $\mathbb{R}^2$ to maximize the number of pairs (p, q) such that $||p - q|| = 1$?*

A strong construction is a $\sqrt{n} \times \sqrt{n}$ grid. Rather than scaling the grid to unit step size, we scale it such that the distance 1 corresponds to a distance d in the integer grid that occurs most frequently.

The possible distances in an unscaled integer grid are $\sqrt{a^2 + b^2}$ for $0 \leq a, b < \sqrt{n}$. By number theory (Landau-Ramanujan), integers representable as a sum of two squares are those where primes $p \equiv 3 \pmod 4$ appear with even exponents.

To get a better bound, we look for "highly composite" integers. Specifically, if we choose a number d composed of a product of many distinct primes $p \equiv 1 \pmod 4$, the number of ways to write d as a sum of two squares is very large. Erdős used this to show there exists a distance occurring:

$$n^{1+\frac{c}{\log\log n}}.$$

This is significantly larger than the $n\sqrt{\log n}$ bound derived from the Pigeonhole Principle on generic integers.

Theorem 3.4 (Upper Bound for Unit Distances)**.** *The number of unit distances is $O(n^{3/2})$.*

Proof. Given n points in $\mathbb{R}^2$, construct a graph where vertices are points and $v \sim u$ if $||v - u|| = 1$. Observe the graph is $K_{2,3}$-free. Geometrically, this is because two circles with radius 1 intersect at most 2 times; thus, no 2 vertices can share 3 common neighbors.
By KST with $s = 2, t = 3$:

$$e(G) \leq \frac{1}{2}\sqrt{3-1} \cdot n^{2-\frac{1}{2}} + o(n^2) = \frac{1}{\sqrt{2}} n^{\frac{3}{2}} + o(n^2).$$

□

3.4 KST for 3-Uniform Hypergraphs

Definition 3.5. *Define $K^{(3)}_{s_1,s_2,t}$ to be the complete 3-uniform, 3-partite hypergraph with parts of size s_1, s_2, t.*

Theorem 3.6 (Generalization of KST to Hypergraphs)**.** *For the 3-uniform case:*

$$ex(n, K^{(3)}_{s_1,s_2,t}) \leq C_{s_1,s_2,t} n^{3-\frac{1}{s_1 s_2}} + O(n^2).$$

Proof. We will count the number of stars $K_{1,1,s_1}$ (sets of edges sharing a common pair of vertices). Let X be this number. For any pair $\{v_1, v_2\}$, let d_{v_1,v_2} be the co-degree (number of edges containing $\{v_1, v_2\}$).

1. **Upper Bound on X:** Fix a set U of size s_1. Define an auxiliary graph G_U on $V(G)$ where $v_1 \sim v_2$ in G_U if $\{v_1, v_2, u\} \in E(G)$ for all $u \in U$. If G_U contains a copy of $K_{s_2,t}$, then combined with U, we form a $K^{(3)}_{s_1,s_2,t}$. Thus, G_U must be $K_{s_2,t}$-free.

$$X \leq \binom{n}{s_1} ex(n, K_{s_2,t}) \leq C \binom{n}{s_1} n^{2-\frac{1}{s_2}}.$$

2. **Lower Bound on X:** Summing over pairs:

$$X = \sum_{\{v_1,v_2\} \in \binom{V}{2}} \binom{d_{v_1,v_2}}{s_1} \geq \binom{n}{2} f_{s_1}\left(\frac{3m}{\binom{n}{2}}\right).$$

Combining bounds (assuming m is large enough):

$$\binom{n}{2} \frac{\left(\frac{3m}{n^2/2}\right)^{s_1}}{s_1!} \lesssim \frac{n^{s_1}}{s_1!} n^{2-\frac{1}{s_2}}.$$

Simplifying:

$$n^2 \left(\frac{m}{n^2}\right)^{s_1} \lesssim n^{s_1+2-\frac{1}{s_2}} \implies m^{s_1} \lesssim n^{3s_1-\frac{1}{s_2}}.$$

Taking the s_1-th root:

$$m \leq C n^{3-\frac{1}{s_1 s_2}}.$$

□

4 Supersaturation, Erdős–Stone, and degenerate Turán problems (Jan 21)

4.1 Supersaturation Lemma

The classical Turán problem asks for the maximum number of edges in a graph avoiding a subgraph F. Supersaturation asks: what happens if we have just *slightly* more edges than this threshold? It turns out we don't just get one copy of F, but exponentially many.

Lemma 4.1 (Supersaturation Lemma)**.** *If G is a graph on n vertices with $e(G) \geq (\pi(F) + \epsilon)\binom{n}{2}$ edges, then G contains at least $\delta(\epsilon)n^{|V(F)|}$ copies of F, where $\delta(\epsilon) > 0$ is a constant independent of n.*

Corollary 4.2. *For the complete graph K_{r+1}, we know $\pi(K_{r+1}) = 1 - \frac{1}{r}$. Thus:*

$$e(G) \geq \left(1 - \frac{1}{r} + \epsilon\right)\binom{n}{2} \implies G \text{ contains } \delta(\epsilon)n^{r+1} \text{ copies of } K_{r+1}.$$

x

4.2 Hypergraph Kővári-Sós-Turán

We generalize the KST theorem from bipartite graphs to r-uniform hypergraphs. We are looking for the Turán number of the complete r-partite hypergraph $K^{(r)}_{s_1,\ldots,s_r}$. Note that the theorem statement often simplifies the last parameter to t.

Theorem 4.3 (Generalized Kővári-Sós-Turán)**.** *Let $K = K^{(r)}_{s_1,s_2,\ldots,s_{r-1},t}$ be the complete r-partite r-uniform hypergraph with part sizes $s_1, \ldots, s_{r-1}, t$. Then:*

$$ex(n, K) = O_{s,t}\left(n^{r - \frac{1}{s_1 s_2 \cdots s_{r-1}}}\right).$$

Proof. We proceed by induction on the uniformity r. The base case $r = 2$ is the standard graph KST theorem. Assume the bound holds for $(r-1)$-uniform hypergraphs.
Let G be an r-uniform hypergraph with n vertices and m edges. We define a "hypergraph star" $K^{(r)}_{1,1,\ldots,s_1}$ as a set of s_1 edges that all share a common core of $r-1$ vertices.
Let X be the number of such stars (copies of $K^{(r)}_{1,1,\ldots,1,s_1}$) in G. We double count X.

1. Lower Bound (Convexity): For every subset $S \in \binom{V(G)}{r-1}$ of size $r-1$, let the *co-degree* d_S be the number of edges in G containing S. A set S contributes $\binom{d_S}{s_1}$ stars to the count.

$$X = \sum_{S \in \binom{V(G)}{r-1}} \binom{d_S}{s_1}.$$

Using Jensen's Inequality (and the convexity of $\binom{x}{s_1}$), we bound this sum:

$$X \geq \binom{n}{r-1}\binom{\bar{d}}{s_1}, \quad \text{where } \bar{d} = \frac{\sum d_S}{\binom{n}{r-1}} = \frac{rm}{\binom{n}{r-1}}.$$

Approximating for large n: $\bar{d} \approx \frac{r!m}{n^{r-1}}$ and $\binom{x}{k} \approx \frac{x^k}{k!}$, we get:

$$X \gtrsim n^{r-1} \cdot \left(\frac{m}{n^{r-1}}\right)^{s_1}. \tag{1}$$

2. Upper Bound (Induction): Fix a set of vertices $V_1 = \{v_1, \dots, v_{s_1}\}$. We ask: how many ways can we complete this into a star?
Define an auxiliary $(r-1)$-uniform hypergraph G_{V_1} on $V(G)$. A set e' of size $r-1$ is an edge in G_{V_1} if $e' \cup \{v_i\} \in E(G)$ for all $i = 1, \dots, s_1$. In other words, e' is in the common neighborhood of all v_i.
If G_{V_1} contains a copy of $K^{(r-1)}_{s_2,\dots,s_{r-1},t}$, then combined with V_1, we would form a copy of the forbidden $K^{(r)}_{s_1,\dots,t}$ in G. Thus, G_{V_1} must be free of $K^{(r-1)}_{s_2,\dots,t}$. By the inductive hypothesis:

$$e(G_{V_1}) \leq C \cdot n^{(r-1)-\frac{1}{s_2 \cdots s_{r-1}}}.$$

Summing over all choices of V_1 (there are $\binom{n}{s_1}$ such sets):

$$X \leq \binom{n}{s_1} \cdot \max_{V_1} e(G_{V_1}) \lesssim n^{s_1} \cdot n^{r-1-\frac{1}{s_2 \cdots s_{r-1}}}. \tag{2}$$

3. Conclusion: Comparing (1) and (2):

$$n^{r-1} \left(\frac{m}{n^{r-1}}\right)^{s_1} \lesssim n^{s_1+r-1-\frac{1}{s_2 \cdots s_{r-1}}}.$$

Rearranging for m:

$$m^{s_1} \lesssim n^{(r-1)s_1} \cdot n^{s_1-\frac{1}{s_2 \cdots s_{r-1}}} = n^{rs_1-\frac{1}{s_2 \cdots s_{r-1}}}.$$

Taking the s_1-th root:

$$m \lesssim n^{r-\frac{1}{s_1 s_2 \cdots s_{r-1}}}.$$

□

4.3 The Erdős-Stone Theorem

Definition 4.4 (Blow-up)**.** *A **blow-up** of a graph H, denoted $H(s)$, is obtained by replacing each vertex $v \in V(H)$ with an independent set I_v of size s, and replacing each edge $(u, v) \in E(H)$ with a complete bipartite graph between I_u and I_v.*
The Turán graph $T_{n,r}$ is essentially a blow-up of K_r with parts of size n/r.

Theorem 4.5 (Erdős-Stone)**.** *If $e(G) \geq \left(1 - \frac{1}{r} + \epsilon\right)\binom{n}{2}$ and n is sufficiently large, then G contains $T_{s,(r+1)}$ (a blow-up of K_{r+1} with parts of size s).*
Consequently, G contains any graph H with $\chi(H) = r+1$ (since any such H is a subgraph of a large enough blow-up of K_{r+1}).

Proof. **Step 1: Find many cliques.** By the Supersaturation Corollary, since the edge density is greater than the Turán threshold for K_{r+1}, G contains many copies of K_{r+1}:

$$\#K_{r+1} \geq \delta n^{r+1}.$$

Step 2: Construct Auxiliary Hypergraph. Define an $(r+1)$-uniform hypergraph $\mathcal{H}$ with vertex set $V(G)$. Let a set of $r+1$ vertices form an edge in $\mathcal{H}$ if and only if they form a K_{r+1} in the original graph G.
From Step 1, we know $e(\mathcal{H}) \geq \delta n^{r+1}$. Since δ is constant, this is a dense hypergraph.
Step 3: Find a dense structure in the hypergraph. We apply the **Hypergraph KST Theorem** to $\mathcal{H}$. Since $\mathcal{H}$ is dense (order n^{r+1}), it exceeds the KST threshold (which is roughly $n^{(r+1)-\epsilon}$). Therefore, $\mathcal{H}$ must contain a copy of the complete $(r+1)$-partite hypergraph $K^{(r+1)}_{s,s,\dots,s}$.
Step 4: Map back to G. Let the parts of this hypergraph copy be $U_1, \dots, U_{r+1}$, each of size s. The definition of the complete hypergraph implies that for *any* choice of vertices $u_1 \in U_1, \dots, u_{r+1} \in U_{r+1}$, the set $\{u_1, \dots, u_{r+1}\}$ is an edge in $\mathcal{H}$.
By definition of $\mathcal{H}$, this means $\{u_1, \dots, u_{r+1}\}$ forms a clique K_{r+1} in G. If every possible tuple forms a clique, then every pair of vertices in distinct parts U_i, U_j must be connected in G.
Thus, $G[U_1 \cup \dots \cup U_{r+1}]$ contains the complete multipartite graph with parts of size s, which is exactly $T_{s(r+1)}$. □

4.4 Degenerate Turán Numbers (Bipartite Graphs)

When H is bipartite ($r = 1$), the Erdős-Stone bound gives $1 - 1/1 = 0$, which is trivial. We rely on KST bounds $ex(n, K_{s,t}) = O(n^{2-1/s})$. We examine the sharpness of these bounds.

Theorem 4.6 (Tightness of KST)**.**

1. ***Case*** $s = 1$ ***(Trees/Forests):***

$$ex(n, K_{1,t}) \leq \frac{1}{2}(t-1)n.$$

This is tight for a graph consisting of disjoint copies of K_t (or a matching if $t = 1$).

2. ***Case*** $s = 2$ ***(C_4-free graphs):*** *We know $ex(n, K_{2,2}) = O(n^{3/2})$. This is tight.*
Construction (Finite Geometry): *Let q be a prime power. Construct a bipartite graph G with parts A and B, where $|A| = |B| = q^2$.*
   - *Identify vertices in A and B with points in the affine plane $\mathbb{F}_q^2$.*
   - *Define edge $(a, b) \sim (x, y)$ if $ax + by = 1$, where $(a, b) \in A$ and $(x, y) \in B$.*

   For a fixed vertex $v = (a, b) \in A$, its neighborhood is the set of solutions (x, y) to the linear equation $ax+by = 1$. This describes a line in $\mathbb{F}_q^2$, which contains q points (unless $(a, b) = (0, 0)$, which we can discard).

$$e(G) \approx q^2 \cdot q = q^3.$$

   Since $n = 2q^2$, we have $q \approx \sqrt{n/2}$, so $e(G) \approx (n/2)^{3/2} = \Theta(n^{3/2})$.
Why is it C_4-free? *A C_4 corresponds to two vertices in A sharing two common neighbors in B. Geometrically, this means two distinct lines intersect at two distinct points. In a plane, two distinct lines intersect at most at one point. Thus, no C_4 exists.*

3. ***Case*** $s = 3$ ***($K_{3,3}$-free):*** *Tightness $ex(n, K_{3,3}) = \Theta(n^{5/3})$ is known.*
Construction Sketch: *Vertices are points in $\mathbb{F}_q^3$. Adjacency is defined by $(x, y, z) \sim (a, b, c)$ if $(x-a)^2+(y-b)^2+(z-c)^2 = 1$. Geometrically, the neighborhood of a point is a unit sphere. If G contained $K_{3,3}$, three vertices in one part would share three common neighbors. This would imply that three unit spheres intersect at three points. In Euclidean geometry, three spheres intersect at most at 2 points (a circle intersects a sphere at 2 points). This logic carries over to $\mathbb{F}_q$ provided -1 is not a square.*

4. ***Case*** $s = 4$ ***($K_{4,4}$-free):*** *It is an* ***open problem*** *whether the KST bound $O(n^{2-1/4})$ is tight for $K_{4,4}$.*

4.5 Extremal Numbers for Trees

Lemma 4.7. *If a graph G has average degree d, it contains a subgraph H with minimum degree $\delta(H) \geq d/2$.*

Proof. Let $n = |V(G)|$. We have $e(G) = \frac{dn}{2}$. Iteratively delete any vertex with degree strictly less than $d/2$. Let S be the set of removed vertices. The number of edges removed is strictly less than $|S| \cdot \frac{d}{2}$. Even if we remove $n - 1$ vertices, the number of edges removed is $< (n-1)\frac{d}{2} < \frac{dn}{2}$. Thus, the edge set cannot be emptied. The process must stop at a non-empty subgraph where every remaining vertex has degree $\geq d/2$. □

Theorem 4.8. *If T is a tree with r edges (so $r + 1$ vertices), then:*

$$ex(n, T) \leq rn.$$

Proof. Suppose $e(G) > rn$. Then the average degree of G is $d(G) = \frac{2e(G)}{n} > 2r$. By the Lemma, there exists a subgraph $H \subseteq G$ with minimum degree $\delta(H) > r$. Since degrees are integers, $\delta(H) \geq r$.
We embed T into H greedily:

1. Order the vertices of T as $v_1, v_2, \ldots, v_{r+1}$ such that each v_i (for $i > 1$) has exactly one neighbor among $\{v_1, \ldots, v_{i-1}\}$. (This is always possible by doing a BFS/DFS from a root).

2. Map v_1 to any vertex $u_1 \in V(H)$.
3. Suppose we have embedded $v_1, \ldots, v_{i-1}$ as $u_1, \ldots, u_{i-1}$. Let v_j be the parent of v_i in the tree ($j < i$).
4. We need to map v_i to a neighbor of u_j in H.
5. The vertex u_j has at least r neighbors in H. We have used at most $i - 1$ vertices so far. Since $i \leq r + 1$, we have used at most r vertices.
6. We only care that u_j has neighbors not already used in the current tree embedding. The degree of u_j is $\geq r$. The number of currently used vertices is $i - 1 \leq r$.
7. The bound requires strict inequality or careful counting. If $\delta(H) \geq r$, u_j has r neighbors. We have used $i - 1$ vertices total. If $i - 1 < r$, there is space. If $i - 1 = r$, we are placing the last vertex. The neighbors of u_j could be occupied by the other $r - 1$ vertices of the tree.
8. However, the conjecture is actually $\frac{1}{2}(r)n$. The bound rn is loose enough. If $e(G) > rn$, avg deg $> 2r$, min deg $\geq r + 1$. Then u_j has $r + 1$ neighbors, but the tree only has $r + 1$ vertices total. At step i, only $i - 1 \leq r$ vertices are occupied. Thus there is always at least one free neighbor for u_j.

$\square$

5 Cycles, girth, and paths (Jan 23)

5.1 Turán Numbers for Cycles and Girth

Definition 5.1. *If $\mathcal{F}$ is a family of graphs, $ex(n, \mathcal{F})$ = maximum number of edges in an n vertex graph G that contains no $F \in \mathcal{F}$.*

Lemma 5.2. $ex(n, \mathcal{F}) = \left(1 - \frac{1}{r} + o(1)\right)\binom{n}{2}$ *if* $r = \min_{F \in \mathcal{F}} \mathcal{X}(F)$

Proof.

$$ex(n, \mathcal{F}) \leq \min_{F \in \mathcal{F}} ex(n, F).$$

By Erdos Stone, we're done. □

Theorem 5.3. *Let* $\mathcal{F} = \{C_3, ..., C_{2k}\}$ *then* $ex(n, \mathcal{F}) \leq n^{1+\frac{1}{k}}$

Lemma 5.4. *G is $\mathcal{F}$ free iff G has girth $\geq 2k + 1$*

Lemma 5.5. *Every graph G contains a subgraph G' that is bipartite with $e(G') \geq \frac{1}{2}e(G)$.*

Proof. Color vertices red or blue uniformly and independent. Then delete the edges that are monochromatic By independence, for any $e \in E(G)$

$$\Pr[e \in E(G')] = \frac{1}{2} \implies E[e(G')] = \frac{1}{2}e(G).$$

□

We can also prove this with the greedy algorithm and will achieve a slightly better bound.

Proof. To prove theorem 5.3:
Suppose G has at least $n^{1+\frac{1}{k}}$, by lemma 5.2 we may replace G by a sub-graph of $\min \deg d \geq n^{1+\frac{1}{k}}$.
We can do a depth first search by picking a root vertex v. Let V_i be the set of vertices at distance i from v. So $V_0 = \{v\}$ Observe that $G[V_0, V_1, ..., V_k]$ forms a tree with root v and V_i being the descendent at at level i else we get a cycle of length at most $2k$. So we have

$$|V_0| + |V_1| + \cdots + |V_k| \geq 1 + d + (d-1)d + d(d-1)^2 + \cdots + d(d-1)^{k-1} = 1 + d^k > n.$$

So, we have a contradiction. □

Theorem 5.6. $ex(n, C_{2k}) \leq c_k n^{1+\frac{1}{k}}$

Theorem 5.7. $ex(n, \{C_4, C_6, ..., C_{2k}\}) \leq 2n^{1+\frac{1}{k}}$

Remark 5.8. *The exponent $1 + \frac{1}{k}$ is sharp if $k = 2, 3, 5$*

5.2 Turán Numbers for Paths

Theorem 5.9. *Let P_k be a path with k edges then*

$$ex(n, P_k) \leq \frac{k}{2}n.$$

Proof. By induction on n it is enough to work with connected graph as we can just split into connected components. Delete vertices of degree $\leq \frac{k}{2}$. If $e(G) > \frac{k}{2}n$ then this also holds after deletions. So WLOG G has minimum degree at least $\frac{k}{2}$.
Let $v_0, v_1, ..., v_r$ be the longest path in G. The neighbors of v_0, v_r are in the path otherwise you can extend the path. If $v_0 \sim v_r$ then $v_0 v_1 ... v_r$ is a cycle and so $v_i v_{i+1} ... v_r v_0 ... v_{i-1}$ is also a path of same length. Thus, the neighbors of v_i are inside the path. So $\{v_0, v_1, ..., v_r\}$ is a connected conponent of G and thus $r \geq k+1$ else $e(G) \leq \frac{(r-1)n}{2}$. Otherwise, if $v_0 \not\sim v_r$ then there exist i such that $v_0 \sim v_{i+1}$ and $v_r \sim v_i$ then we can produce a cycle $v_0 v_{i+1} v_{i+2} ... v_r v_i v_{i-1} ... v_0$ and this is of length r, so it becomes the previous case. By pigeonhole, such a i exist as $\{i : v_0 v_i + 1 \in E(G)\}$ and $\{i : v_r v_i \in E(G)\}$ must intersect as $r \leq k$ by $v_0 v_r \notin E(G)$ $\square$

6 Projective planes and C_{2k}-free graphs (Jan 26)

6.1 Projective Plane over a Field

Definition 6.1. ***Projective Plane over a Field.*** *Let $\mathbb{F}$ be a field.*

1. *Points: The set of lines through 0 in $\mathbb{F}^3$; equivalently, the 1-dimensional subspaces.*
2. *Lines: The set of planes through 0 in $\mathbb{F}^3$; equivalently, the 2-dimensional subspaces.*
3. *Incidence: A point P lies on a line ℓ if $P \subset \ell$.*

Observe:

1. *Any two points determine a unique line.*
2. *Any two lines intersect in a unique point.*
3. *There exist 4 points no three of which are collinear.*

Inside a projective plane over $\mathbb{F}$, there is a copy of the affine plane $\mathbb{F}^2$ embedded as $\{z = 1\}$.
We denote this by $\mathbb{P}^2(\mathbb{F})$.
We say $p \sim \ell$ if $p \in \ell$ for p a point and ℓ a line of $\mathbb{P}^2(\mathbb{F}_q)$.

6.2 C_4-Free Graph from $\mathbb{P}^2(\mathbb{F}_q)$

Remark 6.2. *There are $q^3 - 1$ nonzero vectors in $\mathbb{F}_q^3 \setminus \{0\}$. We identify points of $\mathbb{P}^2(\mathbb{F}_q)$ with equivalence classes of nonzero vectors under*

$$(x, y, z) \sim (\lambda x, \lambda y, \lambda z) \quad \text{for } \lambda \in \mathbb{F}_q^\times.$$

There are $q - 1$ nonzero choices for λ, so the number of points in $\mathbb{P}^2(\mathbb{F}_q)$ is

$$\frac{q^3 - 1}{q - 1} = q^2 + q + 1.$$

By a similar argument, the number of lines is also $q^2 + q + 1$.
Each line contains exactly $q + 1$ points and dually, each point lies on exactly $q + 1$ lines. There is a natural bijection (called a polarity*) between points and lines, preserving incidence: given a point p, its corresponding line is the orthogonal complement $p^\perp$.*

Theorem 6.3. ***Construction of a C_4-free graph:***
We construct an improved C_4-free graph as follows:

- *The vertices are the points of $\mathbb{P}^2(\mathbb{F}_q)$.*
- *Points p_1 and p_2 are adjacent if and only if $p_1 \in p_2^\perp$ (i.e., p_1 lies on the line associated to p_2's orthogonal complement).*

For any pair p_1, p_2, their common neighborhood is

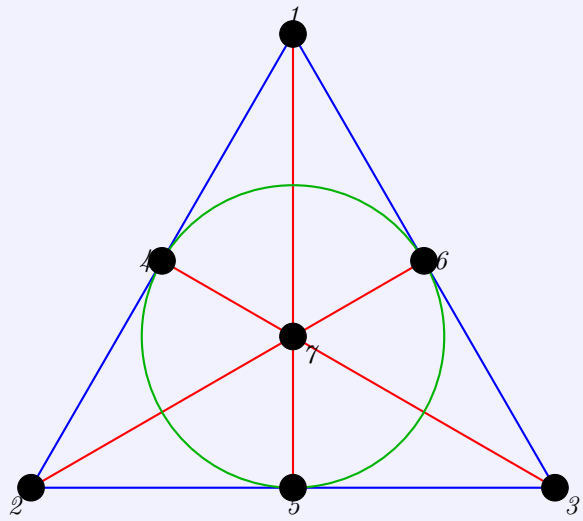
$$p_1^\perp \cap p_2^\perp,$$

which is a single point. Thus, the graph is C_4-free (contains no 4-cycles). This gives $q^2 + q + 1$ vertices with average degree $\approx q$.
Note: There exist p such that $p \subseteq p^\perp$.

6.3 General Projective Planes

Definition 6.4. ***General Projective Planes.*** *If P (points) and L (lines) are sets with an incidence relation such that:*

1. *For any two points, there is a unique line through them.*
2. *Any two lines meet at a unique point.*
3. *There exist four points, no three collinear.*



Example: The Fano plane (order 2) with 7 points and 7 lines. Each line contains 3 points, and each point lies on 3 lines.

Definition 6.5. ***Order of a Projective Plane.*** *The order of a projective plane is the number of points per line minus one (so q if each line has $q+1$ points).*

Remark 6.6. *Are there planes of order which are not prime powers?*

6.4 C_{2k}-Free Graphs for $k = 2, 3, 5$

Theorem 6.7. ***Construction of C_{2k}-free graphs for $k = 2, 3, 5$:***
Let q be a prime power and let $\mathbb{F}_q$ be the finite field with q elements. Define a graph G on the vertex set $\mathbb{F}_q^k$ (the set of k-tuples over $\mathbb{F}_q$) as follows:
For $\mathbf{a} = (a_0, a_1, \ldots, a_{k-1})$ and $\mathbf{b} = (b_0, b_1, \ldots, b_{k-1})$ in $\mathbb{F}_q^k$, declare $\mathbf{a} \sim \mathbf{b}$ if and only if for all $j = 0, 1, \ldots, k-2$,

$$b_j = a_j + a_{j+1} \cdot b_{k-1}$$

where indices are taken modulo k (so $a_k = a_0$).
Then G is C_{2k}-free, i.e., it contains no cycle of length $2k$.

Proof. Suppose for contradiction that G contains a cycle of length $2k$:

$$\mathbf{a}^1, \mathbf{b}^1, \mathbf{a}^2, \mathbf{b}^2, \ldots, \mathbf{a}^k, \mathbf{b}^k,$$

with $\mathbf{a}^i \sim \mathbf{b}^i \sim \mathbf{a}^{i+1}$ for all i, indices modulo k.

Step 1: System of equations. For each $i = 1, \ldots, k$ and $j = 0, \ldots, k-2$, adjacency gives:

$$b_j^i = a_j^i + a_{j+1}^i \cdot b_{k-1}^i \qquad \text{(from } \mathbf{a}^i \sim \mathbf{b}^i\text{)}$$
$$b_j^i = a_j^{i+1} + a_{j+1}^{i+1} \cdot b_{k-1}^i \qquad \text{(from } \mathbf{b}^i \sim \mathbf{a}^{i+1}\text{)}$$

Subtract:

$$(a_j^i - a_j^{i+1}) + (a_{j+1}^i - a_{j+1}^{i+1}) \cdot b_{k-1}^i = 0$$

or

$$(a_j^{i+1} - a_j^i) + (a_{j+1}^{i+1} - a_{j+1}^i) \cdot b_{k-1}^i = 0$$

Summing over i:

$$\sum_{i=1}^{k} (a_j^{i+1} - a_j^i) + \sum_{i=1}^{k} (a_{j+1}^{i+1} - a_{j+1}^i) b_{k-1}^i = 0$$

The first sum telescopes to zero (since indices are cyclic), so for each $j = 0, \ldots, k-2$,

$$\sum_{i=1}^{k} (a_{j+1}^{i+1} - a_{j+1}^i)\, b_{k-1}^i = 0 \tag{3}$$

Step 2: Specialize $j = k-2$. Let $\triangle_i = a_{k-1}^{i+1} - a_{k-1}^i$ and $x_i = b_{k-1}^i$. Then:

$$\sum_{i=1}^{k} \triangle_i x_i = 0$$

and

$$\sum_{i=1}^{k} \triangle_i = 0$$

by telescoping.

Step 3: Linear algebra. The system (3) for $j = 0, \ldots, k-2$ can be written as a $(k-1) \times k$ Vandermonde-type matrix V applied to the vector $(\triangle_1, \ldots, \triangle_k)$:

$$V = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_k \\ x_1^2 & x_2^2 & \cdots & x_k^2 \\ \vdots & \vdots & & \vdots \\ x_1^{k-2} & x_2^{k-2} & \cdots & x_k^{k-2} \end{pmatrix}$$

The equations become $V \cdot (\triangle_1, \ldots, \triangle_k)^\top = 0$. Since $\sum_{i=1}^{k} \triangle_i = 0$, the vector is in the kernel of the row $(1, 1, \ldots, 1)$.

Step 4: Key property of the x_i.

Claim 6.8. *For each i, $x_i \neq x_{i+1}$ (indices modulo k).*

Proof. If $x_i = x_{i+1}$, then $b_{k-1}^i = b_{k-1}^{i+1}$. By the adjacency relations, this forces $\mathbf{b}^i = \mathbf{b}^{i+1}$, so $\mathbf{a}^{i+1} = \mathbf{a}^{i+2}$, contradicting that the cycle is proper. □

Step 5: Contradiction for $k = 2, 3, 5$. Since not all $\triangle_i$ are zero (otherwise the cycle is trivial), and $\sum_{i=1}^{k} \triangle_i = 0$, equation $\sum_{i=1}^{k} \triangle_i x_i = 0$ forces a linear dependence among the x_i (not all distinct). Thus, for these small k we can classify possibilities:

Claim 6.9. *For $k = 2, 3, 5$, if $x_i = x_j$ for some $i \neq j$, then $x_i = x_{i+1}$ for some i, contradicting the above claim.*

- **Case $k = 2$:** Only possible if $x_1 = x_2$, violating $x_1 \neq x_2$.
- **Case $k = 3$:** If $x_1 = x_2$, $x_2 = x_3$, or $x_3 = x_1$, we violate the claim.
- **Case $k = 5$:** If $x_i = x_j$ for $i \neq j$, by relabeling, could assume $x_1 = x_j$ for $j = 2, 3, 4, 5$:
  - $j = 2$ or 5 gives $x_1 = x_2$ or $x_5 = x_1$, directly a violation.
  - $j = 3$: $x_1 = x_3$. The system $\sum_{i=1}^{5} \triangle_i = 0$, $\sum_{i=1}^{5} \triangle_i x_i = 0$ with $x_1 = x_3$ and $x_i \neq x_{i+1}$ for any i only allows further coincidences that eventually force $x_i = x_{i+1}$.

  - $j = 4$: similarly, $x_1 = x_4$ eventually forces a pair $x_i = x_{i+1}$.

- **Case $k = 4$:** The pattern $x_1 = x_3 \neq x_2 = x_4$ is possible with no consecutive equal x_i, which is why this construction fails for $k = 4$.

Therefore, for $k = 2, 3, 5$, no proper $2k$-cycle exists in G. $\square$

7 Algebraic lower bounds for $ex(n, K_{s,t})$ (Jan 28)

7.1 Random Polynomial Construction

Recall: $ex(n, K_{s,t}) \leq c_{s,t} n^{2-\frac{1}{s}} + o(n)$. We will see that $ex(n, K_{s,t}) = \theta\left(n^{2-\frac{1}{s}}\right)$ for $t \geq t_0(s)$.

Theorem 7.1. *Construction of the extremal graph for $ex(n, K_{s,t})$.*

Proof. We will first consider a random graph by picking edges with probability $p = n^{-\frac{1}{s}}$. Let $x^{(1)}, ..., x^{(s)}$ be s vertices chosen uniformly at random without replacement on the left. Let $N(x^{(1),...,x^{(s)}})$ that are the common neighbors with $d(x^{(1)}, ..., x^{(s)}) = |N(...)|$. Let our random variable be $d(x^{(1)}, ..., x^{(s)})$ and is distributed according to

$$Binomial(n, p^s) = Binomial\left(n, \frac{1}{n}\right) \sim Poisson(1).$$

So $\mathbb{E}[d(x^{(1)}, ..., x^{(s)})] = 1$ and $\mathbb{E}[d(...)^k] = o_k(1)$. With probability $\frac{1}{n^\epsilon}$ there is at least $\epsilon \log n$ neighbors. Yet there are $\binom{n}{s}$ such s tuples $(x^{(1)}, ..., x^{(s)})$. and then $\frac{n}{s}$ disjoint. This does not work.

Consider another construction of $K_{s,t}$-free graph with $\Omega(n^{2-\frac{1}{s}})$ edges.
For a bipartite graph with $\mathbb{F}_p^s$ where $n = 2p^s$ on each side we say $x \sim y$ if $f(x, y) = 0$. Pick equation f uniformly at random among all polynomials in s, t of degree at most $d =$ in each of x and y. It would be

$$f = \sum_{|\alpha|,|\beta| \leq d} c_{\alpha\beta} x_1^{\alpha_1} x_2^{\alpha_2} ... x_s^{\alpha_s} y_1^{\beta_1} y_2^{\beta_2} ... y_s^{\beta_s}. \qquad (c_{\alpha\beta} \in \mathbb{F}_p,\ |\alpha| = \alpha_1 + \cdots + \alpha_d)$$

Part I: This random graph is locally (neighborhood look "similar") similar to uniform random graph.
Part II: Globally very different/more discrete. □

Lemma 7.2. *For any fixed $x, y \in \mathbb{F}_p^s$*

$$\Pr[x \sim y] = \frac{1}{p}.$$

Proof. We have

$$f = c_{0,0} + \sum_{\alpha,\beta \neq 0} c_{\alpha\beta} x^\alpha b^\beta.$$

For any choice of $c_{\alpha,\beta}$ fir $(\alpha, \beta) \neq (0, 0)$ there is a unique choice of $c_{0,0}$ such that $f(x, y) = 0$.

$$\mathbb{E}[e(G)] = \frac{1}{p} n^2 = c_s n^{2-\frac{1}{s}}.$$

□

Let $x^{(1)}, ..., x^{(s)}$ be s vertices on say left, $N(x^{(1)}, ..., x^{(s)})$ be the common neighbors with of $x^{(1)}, ..., x^{(s)}$ and this is same as $\{y \in \mathbb{F}_p^s : f(x^{(1)}, y) = 0, ..., f(x^{(s)}, y) = 0\}$.

Lemma 7.3. *If g is a random polynomial of degree of D in m variables and $z^{(1)}, ..., z^{(t)} \in \mathbb{F}_p^m$ then the random variables $g(z^{(1)}), ..., g(z^t)$ are independent if $D > t$ and $p > t^2$.*

Warm up: When $m = 1$, write $g(x) = \underbrace{a_0 + a_1 x + \cdots + a_{t-1} x^{t-1}}_{g_{\text{small}}(x)} + \underbrace{\cdots + a_D x^D}_{g_{\text{large}}(x)}$.

For any choice of $g_{\text{large}}(x)$ and any values $b_1, ..., b_t$ there is a unique polynomial g_{small} such that $g(z^{(i)}) = g_{small}(z^{(i)}) + g_{large}(z^{(i)}) = b_i$ for $i = 1, ..., t$ if and only if $g_{\text{small}}(z^{(i)}) = b_i - g_{large}(z^{(i)})$ by lagrange interpolation.

For the general case, if $z_1^{(1)}, ..., z_1^{(t)}, ..., z_m^{(1)}, ..., z_m^{(t)}$ are distinct, then $g(x_1, ..., x_m) = \overline{g}(x_1) +$ other terms and use $m = 1 =$ case. Pick a invertible linear transform $f : \mathbb{F}_p^s \to \mathbb{F}_p^s$ such that $Tz^{(1)}, ..., Tz^{(t)}$ are

distinct. To pick such a T, let $T(x_1, ..., x_s) = (x_1 + S(x_2, ..., x_s), x_2, ..., x_s)$. For $z^{(i)}, z^{(j)}$ distinct, the space of S for which $T(z^{(i)}) = T(z^{(j)})$ is of co-dimension 1, so a random S has probability $\frac{1}{p}$ of satisfying this. So $\Pr[\exists i, j \text{ such that } T(z^{(i)}) = T(z^{(j)})] \leq \frac{1}{p}\binom{t}{2} < 1$.

We can now conclude from

$$\Pr[f(x^{(1)}, y) = 0, ..., f(x^{(s)}, y) = 0] = \frac{1}{p^s}.$$

If $d > s$ and $p > s^2$ then

$$\mathbb{E}[N(x^{(1)}, ..., x^{(s)})] = 1.$$

Furthermore, if α is a positive integer and $p \cdot s < d$ then

$$\mathbb{E}\left[\left|N(x^{(1)}, ..., x^{(s)})\right|^p\right] = \mathbb{E}\left[\left(\sum_{y \in \mathbb{F}_p^s} R(y)\right)^\alpha\right] = \mathbb{E}\left[\sum_{y^{(1)}, ..., y^{(\alpha)}} R(y^{(1)})...R(y^{(\alpha)})\right].$$

Where $R(y) = \begin{cases} 1 & \text{if } y \in N(x^{(1)}, ..., x^{(s)}) \\ 0 & \text{otherwise} \end{cases}$

$$= \sum_{y^{(1)}, ..., y^{(\alpha)}} p^{-s\#\text{distinct y's}} \leq \sum_k \sum_{y^{(1)}, ..., y^{(k)} \in \mathbb{F}_p^s} p^{-sk} = \sum_k p^{sk} p^{-sk} \cdot \#\text{ways to partition a objection into } k \text{ bins.}$$

Last equality from having to choose $y^{(i)}$ such that k are distinct. This is in $O(1)$. We want to assign values to $y^{(i)}$ such that exactly k are distinct. Equivalently, choose k distinct values in $\mathbb{F}_p^s$. Then choose which of the $y^{(i)}$ go to which bin.

$$\Pr[|N(x^{(1)}, ..., x^{(s)})| \geq T] \leq \frac{\mathbb{E}\left[\left|N(x^{(1)}, ..., x^{(s)})\right|^\alpha\right]}{T^\alpha} = \frac{C_{\alpha,s}}{T^\alpha}.$$

The set $N(x^{(1)}, ..., x^{(s)})$ is contained in $\{y \in \mathbb{F}_p^s : f(x^{(1)}, y) = 0, ..., f(x^{(s)}, y) = 0\}$.
A vector subspace V of $\mathbb{F}_p^s$ of dimension D has p^D elements.

Theorem 7.4. *Lang-Weil Theorem: If $g_1, \ldots, g_r$ are polynomials of degree $\leq D$ in s variables over $\mathbb{F}_p$, then the set $\{y \in \mathbb{F}_p^s : g_1(y) = \cdots = g_r(y) = 0\}$ has either $O_{D,r}(1)$ points, or more than $\frac{p}{2}$ points.*

So $\Pr[\text{some } s \text{ vertices have neighborhood of size} \geq O_{s,\alpha}(1)] \leq C_{s,\alpha}\left(\frac{p}{2}\right)^{-\alpha}\binom{p^s}{s}$. Choose $\alpha = s^2 + 1$. To ensure this probability goes to 0, we need to choose $d = s^2 + 1$.

8 ϵ-regularity for binary words (Feb 4)

8.1 Definitions and Chernoff Bound

Definition 8.1. *Binary word is sequence* $w_1, w_2, ..., w_n \in \{0, 1\}$ *and write* $w = w_1 w_2 ..., w_n$. *The density of* w *is*

$$d(w) = \frac{\#1\text{'s in } w}{n}.$$

Definition 8.2. *Word* w *is* ϵ*-regular if for any subword* w' *of* w *of length* $\geq \epsilon \cdot \mathrm{len}(w)$ *satisfies*

$$|d(w) - d(w')| \leq \epsilon.$$

Lemma 8.3. *By Chernoff bound,*

$$\Pr[|d(w') - d(w)| \geq \epsilon] \leq e^{-c_\epsilon n}.$$

9 Regularity lemma for binary words and twin subsequences (Feb 6)

9.1 The Regularity Lemma

Definition 9.1. *Partition $\mathcal{P}'$ is a refinement of partition $\mathcal{P}$ if each $w' \in \mathcal{P}'$ is a sub-word of some word in $\mathcal{P}$*

Theorem 9.2. *(Regularity Lemma for Binary Words)*
For all $\epsilon > 0$, every word $w \in \{0,1\}^n$ can be partitioned into $\leq M(\epsilon)$ many sub-words $w^{(1)}, ..., w^{(m)}$ (i.e. $w = w^{(1)}...w^{(m)}$). So that the total length of $w^{(i)}$'s that are ϵ-irregular is at most ϵn.

Proof. For a partition $\mathcal{P}$ of w into sub-words $\mathcal{P} = \{w^{(1)}, ..., w^{(m)}\}$ we define $f(\mathcal{P}) = \sum_{w' \in \mathcal{P}} d(w')^2 |w'|/|w|$. Alternatively, we pick a symbol of w at random, say $i \in [n]$ uniformly, then pick another symbol at random from the same part as 1'st, say j is picked.

$$\Pr[w_i = w_j] = \sum_{w' \in \mathcal{P}} \frac{|w'|}{|w|}(d(w')^2 + (1 - d(w'))^2) = 2f(\mathcal{P}) + \sum_{w'} \frac{|w'|}{|w|} - 2\sum_{w'} d(w')\frac{|w'|}{|w|} = 2f(\mathcal{P}) + 1 - 2d(w).$$

Lemma 9.3. *If $\mathcal{P}'$ is a refinement of $\mathcal{P}$ then*

$$f(\mathcal{P}') = f(\mathcal{P}) + \sum_{w' \in \mathcal{P}'} (d(w') - d(\text{parent}(w')))^2 \frac{|w'|}{|w|}.$$

Proof. Expanding the second term we have

$$\sum_{w' \in \mathcal{P}'} d(w')^2 \frac{|w'|}{|w|} - 2d(w')d(\text{parent}(w'))\frac{|w'|}{|w|} + d(\text{parent}(w'))^2 \frac{|w'|}{|w|} = \sum_1 + \sum_2 + \sum_3 .$$

We have $\sum_1 = f(\mathcal{P}')$ and

$$\begin{aligned}
\sum_2 &= \sum_{\tilde{w} \in \mathcal{P}} \sum_{\substack{w' \in \mathcal{P}' \\ \text{parent}(w') = \tilde{w}}} \left(-2d(w')d(\tilde{w})\frac{|w'|}{|w|}\right) \\
&= \sum_{\tilde{w} \in \mathcal{P}} -2d(\tilde{w})\frac{1}{|w|} \sum_{\substack{w' \in \mathcal{P}' \\ \text{parent}(w') = \tilde{w}}} d(w')|w'| \\
&= \sum_{\tilde{w} \in \mathcal{P}} -2d(\tilde{w})\frac{1}{|w|} \cdot d(\tilde{w})|\tilde{w}| \\
&= \sum_{\tilde{w} \in \mathcal{P}} -2d(\tilde{w})^2 \frac{|\tilde{w}|}{|w|} \\
&= -2f(\mathcal{P})
\end{aligned}$$

Then for the third sum

$$\begin{aligned}
\sum_3 &= \sum_{\tilde{w} \in \mathcal{P}} \sum_{\substack{w' \in \mathcal{P}' \\ \text{parent}(w') = \tilde{w}}} d(\tilde{w})^2 \frac{|w'|}{|w|} \\
&= \sum_{\tilde{w} \in \mathcal{P}} d(\tilde{w})^2 \frac{|\tilde{w}|}{|w|} = f(\mathcal{P})
\end{aligned}$$

So we have $\sum_1 + \sum_2 + \sum_3 = f(\mathcal{P}') - f(\mathcal{P})$ □

Start with the trivial partition (i.e. $\mathcal{P} = \{w\}$) and repeat while

$$\sum_{\substack{w' \in \mathcal{P} \\ w' \text{ is } \epsilon\text{-irregular}}} |w'| \leq \epsilon |w|.$$

For $w' \in \mathcal{P}$ that is ϵ-irregular, find a way to write as $w' = w^{(1)}w^{(2)}w^{(3)}$ where $|d(w^{(2)}) - d(w')| \geq \epsilon$ and $|w^{(2)}| \geq \epsilon |w'|$.
Replace w' with the three words $w^{(1)}, w^{(2)}, w^{(3)}$ to obtain a new partition of w, call it $\mathcal{P}'$. Repeat with $\mathcal{P}$ replaced by $\mathcal{P}'$.

To analyze this algorithm, if we drop some words we have

$$f(\mathcal{P}') \geq f(\mathcal{P}) + \sum_{\substack{w' \in \mathcal{P}' \\ w' \text{ is } \epsilon\text{-irregular}}} (d(w^{(2)}) - d(w'))^2 \frac{|w^{(2)}|}{|w|} \geq f(\mathcal{P}) + \sum_{\substack{w' \in \mathcal{P}' \\ w' \text{ is } \epsilon\text{-irregular}}} \epsilon^2 \frac{\epsilon |w'|}{|w|} = f(\mathcal{P}) + \epsilon^4.$$

Note: $f(\mathcal{P}) \leq 1$ for any partition $\mathcal{P}$.
So the number of steps in the algorithm is $\leq \frac{1}{\epsilon^4}$. The theorem holds with $M(\epsilon) = 3^{1/\epsilon^4}$ □

9.2 Twin Subsequences

Example 9.4. *Define a twin in a word $w \in \{0,1\}^n$ is a pair of subsequences $x, y \in w$ that are disjoint (no symbol in both x and y) and are equal as word.*
We define

$$t(w) = \text{the length of longest pair twins in } w \text{ and } t(n) = \min_{w \in \{0,1\}^n} t(w).$$

We have the trivial bounds $\frac{n}{4} \leq t(n) \leq \frac{n}{2}$.

Theorem 9.5.

$$t(n) \geq \left(\frac{1}{2} - o(1)\right) n.$$

Proof. Pick small $\epsilon > 0$ and cut w into ϵ-regular words $t \leq \epsilon n$ junk symbols. Enough $t(w) \geq \left(\frac{1}{2} - c\epsilon\right) |w|$ for ϵ-regular w.
Cut w into $\frac{1}{\epsilon}$ equally long sub-words $w^{(1)}, ..., w^{(m)}$ where $m = \frac{1}{\epsilon}$.
The first twin is 0's from $w^{(1)}$+ 1's from $w^{(2)}$+ 0's from $w^{(3)}$...
The second twin is 1's from $w^{(1)}$+ 0's from $w^{(2)}$+ 1's from $w^{(3)}$...
Choose twins then

$$|\# \text{ 0's in } w^{(i)}| - |\# \text{ 1's in } w^{(2)}| \leq \epsilon |w^{(1)}|.$$

So, # symbols not in either of the twins is less than or equal to

$$\epsilon |w| (\text{1's in } w^{(1)}) + \epsilon |w| (\text{1's in } w^{(2)}) + \sum_i 4\epsilon |w^{(i)}|.$$

□

10 Szemerédi's graph regularity lemma (Feb 13)

10.1 ϵ-Regular Pairs and Partitions

Definition 10.1. *For disjoint sets $U, V \subset V(G)$, the density between U and V is*

$$d(U,V) = \frac{e(U,V)}{|U||V|}.$$

Definition 10.2. *A pair (U,V) of disjoint subsets of $V(G)$ is ϵ-regular if $\forall U' \subseteq U, V' \subseteq V$ such that $|U'| \geq \epsilon|U|$ and $|V'| \geq \epsilon|V|$ we have*

$$|d(U,V) - d(U',V')| \leq \epsilon.$$

Definition 10.3. *We say a partition $V(G) = V_1 \cup ... \cup V_k \cup J$ is ϵ-regular partition when*

1. $|J| \leq \epsilon|V(G)|$
2. $|V_1| = |V_2| = ... = |V_k|$
3. *All except $\leq \epsilon k^2$ pairs (V_i, V_j) are ϵ-regular*

10.2 Proof of the Regularity Lemma

Theorem 10.4. *(Szemeredi's Regularity Lemma) $\forall \epsilon > 0, m$ there exist a constant $M = M(\epsilon, m)$ such that every graph admits an ϵ-regular partition $V(G) = V_1 \cup \cdots \cup V_k \cup J$ where the number of parts $m \leq k \leq M$.*

Remark 10.5. *# edges inside parts + # edges adjacent to J + # edges in ϵ-irregular pairs is at most*

$$k\binom{n/k}{2} + (\epsilon n) \cdot n + \epsilon k^2 (n/k)^2 \leq \frac{1}{k}n^2 + \epsilon n^2 + \epsilon n^2.$$

Proof. For a partition $\mathcal{P}$ of $V := V(G)$, we define the energy as:

$$f(\mathcal{P}) = \sum_{\substack{U,W \in \mathcal{P} \\ U \neq W}} f(U,W) = \sum_{\substack{U,W \in \mathcal{P} \\ U \neq W}} d(U,W)^2 \frac{|U||W|}{|V|^2}$$

Note that because $0 \leq d(U,W) \leq 1$, we trivially have $f(\mathcal{P}) \leq \sum_{U,W} \frac{|U||W|}{|V|^2} \leq 1$ for any partition.

Lemma 10.6. *Suppose $A, B \subset V$ are disjoint and we have partitions $A = A_1 \cup \cdots \cup A_k$ and $B = B_1 \cup \cdots \cup B_l$, then:*

$$\sum_{i,j} f(A_i, B_j) = f(A,B) + \sum_{i,j} (d(A,B) - d(A_i,B_j))^2 \frac{|A_i||B_j|}{|V|^2}$$

(The proof of this lemma follows identically from expanding the variance, as shown in the word version.)

Algorithm Analysis & Energy Increment:
1. The Setup:
We start with an equipartition $\mathcal{P} = \{V_0, V_1, \ldots, V_k\}$ where V_0 is the "Junk" set and $V_1, \ldots, V_k$ are the main parts of equal size. Assume $\mathcal{P}$ is **not** ϵ-regular. By definition, this means the sum of the volume of the ϵ-irregular pairs exceeds ϵ:

$$\sum_{\substack{(V_i,V_j) \\ \text{is } \epsilon\text{-irregular}}} \frac{|V_i||V_j|}{|V|^2} > \epsilon$$

2. Isolating Witnesses:
For each ϵ-irregular pair (V_i, V_j), there exist witness sets $A^{(i,j)} \subset V_i$ and $A^{(j,i)} \subset V_j$ proving the irregularity. They satisfy:

$$|A^{(i,j)}| \geq \epsilon|V_i| \quad \text{and} \quad |A^{(j,i)}| \geq \epsilon|V_j|$$
$$|d(A^{(i,j)}, A^{(j,i)}) - d(V_i, V_j)| \geq \epsilon$$

This creates a boolean partition of V_i into $\{A^{(i,j)}, V_i \setminus A^{(i,j)}\}$ and similarly for V_j.
3. The Common Refinement ($\mathcal{P}_{\text{ref}}$):
For each part V_i, we consider all boolean partitions induced by its irregular pairings. We take the **common refinement** of all these cuts to form a new partition $\mathcal{P}_i$ of V_i.
Since V_i is paired with at most $k-1$ other parts, we cut V_i at most $k-1$ times. Thus, $\mathcal{P}_i$ contains at most $2^{k-1} \leq 2^k$ sub-parts. Let our intermediate refined partition be $\mathcal{P}_{\text{ref}} = \{V_0\} \cup \bigcup_{i=1}^{k} \mathcal{P}_i$.
4. Energy of the Refinement:
We calculate the energy $f(\mathcal{P}_{\text{ref}})$. We can drop non-negative energy terms from pairs of sub-parts that belong to the same parent V_i, and pairs involving the junk set V_0, to establish a strict lower bound:

$$\begin{aligned} f(\mathcal{P}_{\text{ref}}) &= \sum_{\substack{X,Y \in \mathcal{P}_{\text{ref}} \\ X \neq Y}} f(X,Y) \\ &\geq \sum_{1 \leq i < j \leq k} \sum_{\substack{X \in \mathcal{P}_i \\ Y \in \mathcal{P}_j}} f(X,Y) \quad \text{(dropping internal and junk edges)} \end{aligned}$$

Now we apply the Lemma to replace the inner sum. We split the outer sum into regular and irregular pairs:

$$\begin{aligned} f(\mathcal{P}_{\text{ref}}) &\geq \sum_{\text{regular } (i,j)} f(V_i, V_j) + \sum_{\text{irregular } (i,j)} \left[f(V_i, V_j) + \sum_{\substack{X \in \mathcal{P}_i \\ Y \in \mathcal{P}_j}} (d(X,Y) - d(V_i, V_j))^2 \frac{|X||Y|}{|V|^2} \right] \\ &= f(\mathcal{P}_{\text{main}}) + \sum_{\text{irregular } (i,j)} \sum_{\substack{X \in \mathcal{P}_i \\ Y \in \mathcal{P}_j}} (d(X,Y) - d(V_i, V_j))^2 \frac{|X||Y|}{|V|^2} \end{aligned}$$

Because $\mathcal{P}_i$ and $\mathcal{P}_j$ are refinements of the witness sets $A^{(i,j)}$ and $A^{(j,i)}$, we can drop the variance of the non-witness components to strictly bound the variance of the irregular pairs:

$$\begin{aligned} f(\mathcal{P}_{\text{ref}}) &\geq f(\mathcal{P}_{\text{main}}) + \sum_{\text{irregular } (i,j)} (d(A^{(i,j)}, A^{(j,i)}) - d(V_i, V_j))^2 \frac{|A^{(i,j)}||A^{(j,i)}|}{|V|^2} \\ &\geq f(\mathcal{P}_{\text{main}}) + \sum_{\text{irregular } (i,j)} (\epsilon)^2 \frac{(\epsilon|V_i|)(\epsilon|V_j|)}{|V|^2} \\ &= f(\mathcal{P}_{\text{main}}) + \epsilon^4 \sum_{\text{irregular } (i,j)} \frac{|V_i||V_j|}{|V|^2} \\ &> f(\mathcal{P}_{\text{main}}) + \epsilon^4(\epsilon) \quad \text{(applying the irregularity assumption)} \\ &= f(\mathcal{P}_{\text{main}}) + \epsilon^5 \end{aligned}$$

5. Restoring the Equipartition ($\mathcal{P}_{\text{new}}$):
The partition $\mathcal{P}_{\text{ref}}$ has at most $k \cdot 2^k$ parts, but they are not equal sizes. To fix this, we chop every part in $\mathcal{P}_{\text{ref}}$ into equally sized chunks of size $d = \lfloor |V|/(k \cdot 4^k) \rfloor$.
The maximum number of these new parts is $k \cdot 4^k$.
For each part in $\mathcal{P}_{\text{ref}}$, the leftover vertices (which number at most d) are thrown into the new junk set.
The number of parts in $\mathcal{P}_{\text{ref}}$ is $\leq k2^k$, so the total number of vertices added to the junk is bounded by:

$$\text{New Junk Added} \leq (k2^k) \cdot d \leq (k2^k)\frac{|V|}{k4^k} = \frac{|V|}{2^k}$$

Let this new equipartition be $\mathcal{P}_{\text{new}}$. Because chopping parts smaller only refines the partition further, the energy cannot decrease. Thus, the energy strictly increases by our previous bound:

$$f(\mathcal{P}_{\text{new}}) \geq f(\mathcal{P}_{\text{ref}}) > f(\mathcal{P}_{\text{main}}) + \epsilon^5$$

6. Final Iteration Bound:
Because the energy is bounded by 1, and each step strictly increases the energy by at least ϵ^5, the algorithm must terminate in at most $T = \lceil 1/\epsilon^5 \rceil$ steps.
Let m be the initial number of parts. At each step, if we have k parts, the next step yields at most $g(k) = k \cdot 4^k \leq 4^{4^k}$ parts.
The total number of parts at termination is bounded by iterating this function T times:

$$M \leq g^{(T)}(m) \leq \underbrace{4^{4^{4^{\cdots 4^m}}}}_{2/\epsilon^5 \text{ times}}$$

The total accumulated junk size over all iterations is bounded by $\sum_{k \geq m} \frac{|V|}{2^k} \leq \frac{|V|}{2^{m-1}}$. By picking our initial m large enough (e.g., $2^{m-1} \geq 1/\epsilon$), we guarantee the final junk size remains $\leq \epsilon|V|$. This completes the proof. $\square$

11 Triangle counting and removal lemmas (Feb 16)

11.1 Triangle Counting Lemma

Lemma 11.1 (Triangle Counting Lemma). *Let $X, Y, Z \subset V(G)$ be disjoint with $d(X,Y), d(X,Z), d(Y,Z) \geq 2\epsilon$, and suppose (X,Y), (X,Z), (Y,Z) are ϵ-regular.*
Then the number of XYZ-triangles is at least

$$(1-2\epsilon)(d(X,Y)-\epsilon)(d(X,Z)-\epsilon)(d(Y,Z)-\epsilon)|X||Y||Z|$$

Proof. Observe the following:

1. $Y_{\text{bad}} = \{x \in X : \deg_Y(x) < (d(X,Y)-\epsilon)|Y|\}$
2. $Z_{\text{bad}} = \{x \in X : \deg_Z(x) < (d(X,Z)-\epsilon)|Z|\}$

Then $|Y_{\text{bad}}|, |Z_{\text{bad}}| \leq \epsilon|X|$.
Consider Y_{bad},

$$e(Y_{\text{bad}}, Y) = \sum_{x \in Y_{\text{bad}}} \deg_Y(x) < |Y_{\text{bad}}|(d(X,Y)-\epsilon)|Y|.$$

By ϵ-regularity we have $d(X,Y)-\epsilon$ is within ϵ of $d(X,Y)$ so a contradiction if $|Y_{\text{bad}}| > \epsilon|X|$.

To prove the Triangle Counting Lemma, let $X' := X \setminus (Y_{\text{bad}} \cup Z_{\text{bad}})$ then $|X'| \geq (1-2\epsilon)|X|$ by observation.
Let $x \in X'$ then $N_Y(x) := N(x) \cap Y$ and $N_Z(x) := N(x) \cap Z$.
The number of triangles containing x is

$$e(N_Y(x), N_Z(x)) \geq |N_Y(x)||N_Z(x)|(d(Y,Z)-\epsilon). \tag{$*$}$$

By ϵ-regularity of (Y,Z) and definition of bad. Additionally

$$(*) \geq (d(X,Y)-\epsilon)(d(X,Z)-\epsilon)(d(Y,Z)-\epsilon)|Y||Z|.$$

So

$$\#\triangle = \sum_{x \in X'} (*) \geq (*)(1-2\epsilon)|X|.$$

□

11.2 Triangle Removal Lemma

Lemma 11.2. *Triangle Removal Lemma: $\forall \epsilon > 0, \exists \delta > 0$ such that every graph with $\leq \delta n^3$ triangles can be made triangle free by deleting only ϵn^2 edges.*

Proof. Pick $\epsilon' = \frac{\epsilon}{4}$ and $m = \frac{4}{\epsilon}$ and apply Szemeredi's Regularity Lemma to G to get an ϵ-regular partition $V(G) = V_1 \cup \cdots \cup V_k \cup J$ where $m \leq k \leq M(\epsilon', m)$ and $|J| \leq \epsilon' n$. To clean up the graph

1. Delete all edges incident to J as we do not know anything about J
2. Delete all edges inside ϵ-irregular parts
3. Delete all edges inside a single V_i
4. Delete edges between V_i and V_j if $d(V_i, V_j) < 2\epsilon'$

If at least one $\triangle$ survives, say xyz with $x \in X, y \in Y, z \in Z$ with $X, Y, Z \in \{V_1, ..., V_k\}$ then X, Y, Z satisfy the condition of triangle counting lemma. So we have

$$\#\triangle \in G \geq \#XYZ\triangle \geq (1-2\epsilon')(d(X,Y)-\epsilon)(d(X,Z)-\epsilon)(d(Y,Z)-\epsilon)|X||Y||Z| \geq (1-2\epsilon)\epsilon^3 \left(\frac{|V(G)|(1-\epsilon')}{M}\right)^3.$$

So the total # edges deleted in steps (1)-(4) is at most

$$n|J| + \#\epsilon'\text{-irregular pairs} \cdot \left(\frac{n}{k}\right)^2 + k\binom{\frac{n}{k}}{2} + k^2 \cdot \epsilon' \left(\frac{n}{k}\right)^2 \leq \epsilon' n^2 + \epsilon' k^2 \left(\frac{n}{k}\right)^2 + \frac{n^2}{m} + \epsilon' n^2.$$

So the bound we get is $\delta \geq \frac{1}{\text{tower}(\frac{c}{\epsilon^5})}$ $\square$

12 Roth's theorem and AP-free sets (Feb 18)

12.1 Statements: Roth, Szemerédi, and the corner theorem

Theorem 12.1. *Roth's Theorem (1953): If $A \subset [n]$ of density at least $\alpha > 0$ (i.e. $|A| \geq \alpha n$) and $n \geq n_0(\alpha)$ then A contains a 3-term arithmetic progression i.e., a set of the form $x, x+d, x+2d$ with $d \neq 0$*

Remark 12.2. *Same holds for $A \subset P$ for any n-term arithmetic progression P.*

Theorem 12.3. *Szemeredi Theorem (1974): $\forall \alpha > 0, \forall k$ if $n \geq n_0(\alpha, k)$ then $|A| \geq \alpha n \implies A$ contains a k-term arithmetic progression.*

12.2 The corner theorem

Theorem 12.4. *Corner Theorem: $\forall \beta > 0$ and $A \subset [n]^2$ with $|A| \geq \beta n^2$ and $n \geq n_0(\beta)$ then A contains a "corner" which is a set of the form $(x, y), (x+d, y), (x, y+d)$ with $d \neq 0$*

Remark 12.5. *Corner Theorem $\implies$ Roth's Theorem: Given $A \subset [n]$ define $B \subset [2n]^2$ by $B = \{(x, y) \in [2n]^2 : x - y \in A\}$. We have*

$$|B| \geq n|A| \geq \alpha n^2.$$

So with $\beta = \frac{\alpha}{4}$ and applying corner's theorem and projecion of $B \to A$ we have a 3-term arithmetic progression.

Proof. Corner Theorem: Define a tripartite graph with parts V, H, D where the vertices of V are all vertical lines through $[n]^2$ and the vertices of H are all horizontal lines through $[n]^2$ and the vertices of D are all diagonal lines through $[n]^2$. Then we have

$$|V| = |H| = |D| = n + n + (2n - 1) \leq 4n.$$

Put an edge between lines ℓ, ℓ' if $\ell \cap \ell' \in A$. Each point of A implies there are 3 edges of G that form a $\triangle$ in G. Furthermore, these $\triangle$'s are edge disjoint.
Edge disjointness implies to kill all $\triangle$'s in G we need to delete at least $|A| = \beta n^2$ edges.
By triangle removal the number of $\triangle$'s in G is at least $\delta(\beta)n^3$.
The number of non-degenerate $\triangle$'s is $\geq \delta n^3 - n^2 > 0$. The intersection points of a non-degenerate $\triangle$ form a corner. □

12.3 Lower bounds: greedy and Behrend

Theorem 12.6. *Construction of such a set by Greedy Algorithm:*

$$\{m : \textit{all digits of } m \textit{ in base 3 are 0 or 1}\}.$$

There are no 3-term arithmetic progression in this set so we have $|A| = n^{\log_3 2}$.

Theorem 12.7. *(Behrend Construction): Define $r_3(n) = \max\{|A| : A \subseteq [n], A \textit{ is 3 AP-free}\}$. We have $r_3(N) \geq Ne^{-c\sqrt{\log N}}$.*

Proof. Let $n =$ and consider $[n]^d \subset \mathbb{R}^d$.
Note: Any sphere in $\mathbb{R}^d$ is a 3-AP-free set as a line intersects a line in at most 2 points.
For $x \in [n]^d$, $||x||^2 \leq dn^2$. So by pigeonhole principle there exist R such that $S = \{x \in [n]^d : ||x||^2 = R\}$ is of size $|S| \geq \frac{n^d}{dn^2} = \frac{n^{d-2}}{d}$

Treat elements of $[n]^d$ as numbers in base $2n$ of length d. So $r_3(\underbrace{(2n)^d}_{N}) \geq \frac{n^{d-2}}{d} = \frac{(2n)^d}{2^d nd}$.

Let $d = c\sqrt{\log N}$. Given N choose d such that we minimize $2^d nd$. $\square$

13 Regularity applications: $K_{t,t,t}$, counting F-free graphs, and 3-graphs (Feb 20)

13.1 Embedding $K_{t,t,t}$ from regular triples

Theorem 13.1. *Sligth Generalization of Triangle Counting Lemma: If X, Y, Z are disjoint subsets of $V(G), (X,Y), (X,Z), (Y,Z)$ are ϵ-regular then G contains $K_{t,t,t}$, $d(X,Y), d(X,Z), d(Y,Z) \geq \alpha$ and $\epsilon < \epsilon_0(\alpha, t)$.*

Proof. Sketch: Embed one vertex at a time. Originally no vertices are embedded, $X_0 = X, Y_0 = Y, Z_0 = Z$. Pick some unembedded vertex v of G, say v needs to be embedded in X. Pick $x \in X_i$ such that $N_Y(x), N_Z(x)$ are large. □

13.2 Erdős–Stone for chromatic number 3

Theorem 13.2. *Erdos Stone Theorem for graphs F with $\mathcal{F} = 3$, (i.e. $ex(n,F) = \left(\frac{1}{2} + o(1)\right) n^2$)*

Proof. Given F-free G, make a ϵ-reduced graph R with # edges deleted is $\leq \epsilon' n^2$ where $\epsilon' \to 0$.
If reduced graph contains $\triangle$, then G contains $K_{t,t,t}$ with $t = |V(F)|$ implying G contains F. So, the reduced graph is $\triangle$-free. By Turan's Theorem for $\triangle$'s,

$$e(R) \leq ex(k, \triangle) = \frac{\lfloor k^2 \rfloor 2}{=} \left(\frac{1}{2} + o(1)\right) \binom{k}{2}.$$

where $V(R) = \{V_1, ..., V_k\}$.

$$e(G) \leq \epsilon' n^2 + \sum_{V_i \sim V_j} e(V_i, V_j) \leq \epsilon' n^2 + \left(\frac{1}{2} + o(1)\right) \binom{k}{2} \left(\frac{n}{k}\right)^2.$$

In the regularity, we can choose the lower bound on number of parts, and so we can ensure $k \to \infty$ with $|V(G)|$. □

13.3 Counting F-free graphs (chromatic number 3)

Theorem 13.3. *Let F be a graph with $(F) = 3$ then the number of F-free graphs on $[n]$ is*

$$2^{ex(n,F)(1+o(1))}.$$

Proof. Given F-free graph G, apply Regularity Lemma and clean it up to obtain G' whose reduced graph is R where $R = R(G)$, $G' = G'(G)$ and $k = |V(R)|$ also depends on G and satisfies

1. $k \leq M(\epsilon)$
2. $E(G') \subseteq E(G)$
3. $|E(G) \setminus E(G')| \leq \epsilon n^2$
4. R is triangle free

For each choice of R there are

$$\leq \underbrace{\binom{\binom{n}{2}}{\epsilon n^2}}_{\substack{\text{deleted}\\ \text{edges}}} \cdot \underbrace{(2^n)^k}_{\substack{\text{choices for}\\ V_1,\ldots,V_k}} \cdot \underbrace{2^{(n/k)^2 \cdot e(R)}}_{\substack{\text{choices of edges}\\ \text{between regular parts}}}.$$

Note that

$$2^{(n/k)^2 e(R)} \leq 2^{(n/k)^2 \cdot \lfloor \frac{k^2}{2} \rfloor} = 2^{\left(\frac{1}{4} + o(1)\right) n^2}.$$

where $o(1) \to 0$ as $k \to \infty$.

$$\binom{\binom{n}{2}}{\epsilon n^2} \leq 2^{H(\epsilon)n^2}.$$

where $H(\epsilon) = \log_2\left(\frac{1}{\epsilon}\right) + (1-\epsilon)\log_2\left(\frac{1}{1-\epsilon}\right)$.
To bound the number of R, it is at most the number of graphs on $\leq M(\epsilon)$ vertices $\leq M'(\epsilon)$.
The $\#F$ free graphs is at most

$$2^{(1/4+o(1))n^2} \cdot 2^{H(\epsilon)n^2} \cdot 2^{nM(\epsilon)} M'(\epsilon).$$

□

13.4 3-uniform hypergraph regularity (definition)

Definition 13.4. *Hypergraph Regularity Lemma: For $H \subseteq \binom{[n]}{3}$ is a 3-uniform hypergraph with ϵ-regularity. Triples (X, Y, Z) with $X, Y, Z \subseteq V$ is ϵ-regular if*

$$\left| \frac{e(X,Y,Z)}{|X||Y||Z|} - \frac{e(X',Y',Z')}{|X'||Y'||Z'|} \right| \leq \epsilon.$$

For all $|X'| \geq \epsilon|X|$, $|Y'| \geq \epsilon|Y|$, $|Z'| \geq \epsilon|Z|$.

Example 13.5. *Each edge in X, Y, Z is chosen independently with probability p.*

Example 13.6. *Pick a random tripartite graph (=2-uniform hypergraph) on X, Y, Z with edges chosen with probability p. Call this graph G define $H = \{\{x, y, z\} : x \sim y, x \sim z, y \sim z\}$.*

14 Extremal set theory: Sperner and intersecting families (Feb 23)

14.1 Antichains and Sperner's theorem

Definition 14.1. *If $(P, \succeq)$ is a poset then $A \subseteq P$ is a chain if for all $a, b \in A$ does it hold that $a \subseteq b$.*

Definition 14.2. *A family of $\mathcal{F} \subset 2^{[n]}$ is intersecting if an antichain if for no pair $A, B \in \mathcal{F}$ does it hold that $A \subset B$.*

Theorem 14.3. *(Sperner's Theorem) If $\mathcal{F} \subset 2^{[n]}$ is an intersecting family then $|\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$.*

Definition 14.4. *For a family $\mathcal{A} \subset \binom{[n]}{k}$ it's shadow is*

$$\partial\mathcal{A} = \{\mathcal{S} \in \binom{[n]}{k-1} : \mathcal{S} \cup \{i\} \in \mathcal{A} \text{ for some } i\}.$$

Lemma 14.5. *If $\mathcal{A} \subseteq \binom{[n]}{k}$ then $\partial\mathcal{A} \geq \frac{k}{n-k}|\mathcal{A}|$*

Proof. Let $X = \{(A', A) \in \partial\mathcal{A} \times \mathcal{A} : A' \subset A\}$ then be the number of edges in the bipartite graph between $\mathcal{A}$ and $\mathcal{A}'$. We have $|X| = |\mathcal{A}| \cdot k$ and $|X| \leq |\partial\mathcal{A}|(n-k+1)$. So we have

$$|\partial\mathcal{A}| \geq \frac{k}{n-k}|\mathcal{A}|.$$

□

Proof. To show Sperner's Theorem: Suppose $\mathcal{F}$ is an antichain with $a = \min\{|A| : A \in \mathcal{F}\}$ and $b = \max\{|A| : A \in \mathcal{F}\}$.
Induction on $b + (\frac{m}{2} - a)$. (Induction Step) If $b > \frac{n}{2}$ then $\mathcal{A} = \mathcal{F} \cap \binom{[n]}{b}$ and set $\mathcal{F}' = (\mathcal{A} \setminus \partial\mathcal{A}) \cup \partial\mathcal{A}$.
Note that $\mathcal{F}'$ is an antichain.
If $x, y \in \mathcal{F} \setminus \mathcal{A}$ then x, y are incomparable since $x, y \in \mathcal{F}$.
If $x, y \in \partial\mathcal{A}$, then x, y are incomparable since they are in same level.
If $x \in \mathcal{F} \setminus \mathcal{A}$ and $y \in \partial\mathcal{A}$ then $|X| \leq |Y|$. So if X and Y were comparable then $X \subset Y$. So if $Y \in \partial\mathcal{A}$ then $\exists Z \in \mathcal{A}, Y \subset Z \implies x \subset Y \subset Z$. A contradiction.
Then calculating the size of $|\mathcal{F}'| = |\mathcal{F}| - |\mathcal{A}| + |\partial A|$ then

$$|\partial\mathcal{A}| \geq \frac{b}{n-b+1}|\mathcal{A}| \geq \frac{(n+1)/2}{(n+1)/2}|\mathcal{A}|.$$

If $a < \frac{n-1}{2}$, upshadow or replace $\mathcal{F}$ by $\{[n] \setminus A : A \in \mathcal{F}\}$ and use down shadow. □

Proof. Second proof: Picka random maximal chain in $2^{[n]}$

$$\emptyset = A_0 \subset A_1 \subset \cdots \subset A_n = [n].$$

Where $|A_i| = i$ uniformly at random among all such chain. Equivalently, pick a random permutation $\pi \in S_n$ and let $A_i = \{\pi(1), ..., \pi(i)\}$.
Let $A = \{A_0, ..., A_n\}$ then $|A \cap F| \leq 1$.

$$1 \geq \mathbb{E}[|A \cap F|] = \sum_{A \in \mathcal{F}} \Pr[A \in \mathcal{A}] = \sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \geq \frac{|\mathcal{F}|}{\binom{n}{\lfloor n/2 \rfloor}}.$$

□

14.2 Intersecting families

Definition 14.6. *Family $\mathcal{F} \subset 2^{[n]}$ is intersecting if $\forall A, B \in \mathcal{F}$, $A \cap B \neq \emptyset$.*

Theorem 14.7. *If $\mathcal{F}$ is intersecting, $\mathcal{F} \subset 2^{[n]}$ then $|\mathcal{F}| \leq 2^{n-1}$.*

Proof. For each S at most one of $S, [n] \setminus S$ can be in $\mathcal{F}$. □

14.3 Erdős–Ko–Rado for fixed k

Theorem 14.8. *If $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting then*

$$|\mathcal{F}| \leq \begin{cases} \binom{n-1}{k-1} & \text{if } k \leq \frac{n}{2} \\ \binom{n}{k} & \text{if } k > \frac{n}{2} \end{cases}.$$

Proof. Pick a random permutation $\pi \in S_n$ and put elements $\pi(1), ..., \pi(k)$ on a circle. Let $|\mathcal{A}|$ be all k-element sets made of consecutive elements on the circle.
Let $\mathcal{A}$ be all k-element sets made of consecutive elements on the circle.

$$\mathcal{A} := \{\{\pi(i), \pi(i+1), ..., \pi(i+k-1)\} : i \in [n]\}.$$

Proposition 14.9. *For any $\pi \in S_n$, $|\mathcal{A} \cap \mathcal{F}| \leq k$*

WLOG $\pi = (1\ 2\ ...n)$, consider any set $S \in \mathcal{A} \cap \mathcal{F}$ (if there are none we're done). By circular symmetry (replacing S by $S + \Delta \pmod n$).
WLOG $S = \{k, k+1, ..., 2k-1\}$ and every other $S' \in \mathcal{A} \cap \mathcal{F}$ intersect S. The intervals that intersect S are

$$\{1, 2, ..., k\}, \{2, 3, ..., k+1\}, ..., \{n-k+1, n-k+2, ..., n\}.$$

$$\{k+1, k+2, ..., 2k\}, \{k+2, ..., 2k+1\}, ..., 2k_1, 2k, ..., 3k-2.$$

On each line the two intervals are disjoint and so at most one is in $\mathcal{A} \cap \mathcal{F}$. So there are at most k intervals.

$$\mathbb{E}[|\mathcal{A} \cap \mathcal{F}|] = \sum_{A \in \mathcal{F}} \Pr[A \in \mathcal{A}] = \frac{|\mathcal{F}|}{\binom{n}{k}} \cdot n$$

$$\Downarrow$$

$$n \leq k \implies |\mathcal{F}| \leq \binom{n-1}{k-1}$$

Note: The probability is from $\forall s \in \mathcal{F}$, $\forall i \in [n]$,

$$\Pr[\{\pi(i), \pi(i+1), ..., \pi(i+k-1)\} = S] = \frac{1}{\binom{n}{k}}.$$

□

15 t-intersecting hypergraphs and Radon's theorem (Mar 9)

15.1 A Frankl-type bound for t-intersections

Theorem 15.1. *Let H be k-uniform hypergraph and $\forall h_1, h_2 \in H$, $|h_1 \cap h_2| \geq t$ then $\exists n_0(k,t)$ such that if $n \geq n_0(k,t)$ and $n = |V(H)|$ then*

$$|H| \leq \binom{n-t}{k-t}.$$

Moreover, $|H| = \binom{n-t}{k-t}$ implies $\exists x_1, ..., x_t$ such that $x_i \in h$ for all $i \in [t]$ and $h \in H$.

Proof. We want to start by choosing $A, b \in H$ such that $|A \cap B| = t$.
The other case is that is is $t+1$ intersecting. Fix A then for every other $B \in H$ has $\geq t+1$ points in A.
So

$$\begin{aligned}
|H| &\leq 1 + \binom{k}{t+1}\binom{n-k}{k-t-1} + \binom{k}{t+2}\binom{n-k}{k-t-2} + ... + \binom{k}{k}\binom{n-k}{k-k} \\
&\leq C_{k,t}\binom{n-k}{k-t-1} \\
&< \binom{n-k}{k-t}
\end{aligned}$$

So we're done with this case.
If every other $C \in H$ contains t points of $A \cap B$ we're done.
Otherwise, $\exists C$ such that $C \not\supseteq A \cap B$. Such a C must contain t points from $A \setminus B$ and $B \setminus A$.
We claim that every other edge $D \in H$ intersects $(A \cup B \cup C)$ in $\geq t+1$ points. This is easily shown by casework.
Thus

$$\begin{aligned}
|H| &\leq 3 + \binom{3k}{t+1}\binom{n-3k}{k-t-1} \\
&\leq 3 + \sum_{t+1 \leq \ell \leq k \leq j \leq 3k} \binom{j}{\ell}\binom{n-j}{k-\ell} \\
&\leq C_{k,t}\binom{n}{k-t-1} < \binom{n-t}{k-t}
\end{aligned}$$

□

15.2 Affine dependence and Radon's theorem

Definition 15.2. *Affine Dependence: Given $p_1, ..., p_n \in \mathbb{R}^d$, they are affinely dependent iff $p_1 - p_n, p_2 - p_n, ..., p_{n-1} - p_n$ are linearly dependent iff $\exists a_1, ..., a_{n-1}$ such that*

$$a_1(p_1 - p_n) + ... + a_{n-1}(p_{n-1} - p_n) = 0.$$

Equivalently,

$$a_1 p_1 + ... + a_{n-1}p_{n-1} + (-a_1 - ... - a_{n-1})p_n = 0.$$

An equivalent definition derived from this would be $\exists a_1, ..., a_{n-1}$ such that $\sum a_i p_i = 0$ and $\sum a_i = 0$.

Theorem 15.3. *Rudon's Theorem: Given $d+2$ points in $\mathbb{R}^d$, $p_1, ..., p_{d+2}$, the points can be red or blue colored such that Conv(Red)$\cap$Conv(Blue) is nonempty.*

Proof. $p_1, ..., p_{d+2}$ are affinely dependent since $d+2 \geq d+1$ so $\exists a_1, ..., a_{d+2}$ such that $\sum a_i p_i = 0$ and $\sum a_i = 0$.
We color p_i red if $a_i \geq 0$ and blue otherwise.

WLOG let $p_1, ...p_k$ be red and $p_{k+1}, ..., p_{d+2}$ be blue.
Letting $b_i = -a_i$ for $i = k+1, ..., d+2$ we have

$$a_1p_1 + ... + a_kp_k - b_{k+1}p_{k+1} - ... - b_{d+2}p_{d+2} = 0. \tag{1}$$

Let $A = a_1 + ... + a_k$ then $\frac{a_1}{A}p_1 + \cdots + \frac{a_k}{A}p_k$ is a convex combination of red points so it's in the convex hull of the red points. We now want to show that this is also in the convex hull of the blue points.
From (1) we have that

$$\frac{a_1}{A}p_1 + \cdots + \frac{a_k}{A}p_k = \frac{b_{k+1}}{A}p_{k+1} + \cdots + \frac{b_{d+2}}{A}p_{d+2}.$$

This is again a convex combination of blue points as $\sum a_i = 0$. $\square$

16 Helly, Carathéodory, and transversals (Mar 11)

16.1 Helly's theorem

Theorem 16.1. *Helly's Theorem: In $\mathbb{R}^d$ for $m \geq d+1$, if convex sets $A_1, ..., A_m \in \mathbb{R}^d$ have the property that any $d+1$ sets have nonempty intersection then $A_1 \cap \cdots \cap A_n$ is nonempty.*

Remark 16.2. *$d+1$ is necessary. Consider $E_1, ..., E_d$ where $E_i = \{x \in \mathbb{R}^d | x \cdot e_i \geq 0\}$ and $E_{d+1} = \{x \in \mathbb{R}^d | \sum x_i \leq -1\}$*

Proof. (Base Case): $m = d+2$, given $H_1, ..., H_{d+2}$ with any $d+1$ intersect and there's $\binom{d+2}{d+1} = d+2$ such intersections, take a point $p_i \in \bigcap_{1 \leq j \leq d+2, j \neq i} H_j$. Radon's theorem gives red/blue coloring of the points such that $\exists q \in CH(Red) \cap CH(Blue)$.
Claim: $q \in H_i$ for all i.
WLOG we'll show $q \in H_1$ and WLOG assume $p_1 \in \bigcap_{2 \leq j \leq m} H_j$ got colored blue by Radon. Then all the red points $p_j \in$Red are in H_1. Since H_i is convex, $q \in CH(Red) \subseteq H_1$.
(Induction Step): Given $H_1, ..., H_m$. Given $m \geq d+2$ sets, any $d+1$ intersect. By base case, any $d+2$ intersect. Define $H'_1 = H_1, H'_2 = H_2, ..., H'_{m-2} = H_{m-2}$ and $H'_{m-1} = H_{m-1} \cap H_m$. After doing this, still any $d+1$ intersect so we're done by induction. □

Theorem 16.3. *If $H_1, H_2, ...$ are convex, compact sets and any $d+1$ intersect, they all intersect.*

Proof. X compact implies if every finite intersection of closed sets nonempty, then every intersection of closed set nonempty. □

Theorem 16.4. *A collection of vertical line segments has a transversal iff any three of them do.*

Proof. Let the convex sets $H_1, ..., H_n$ be defined as H_i being the set of lines intersecting ℓ_i. Done by Helly's Theorem. □

16.2 Carathéodory's theorem

Theorem 16.5. *Caratheodory's Theorem: In $\mathbb{R}^d$, any convex set C can be written as the convex hull of its $d+1$ points.*

Proof. Assume WLOG $0 \in CH(x_1, ..., x_m)$ but not in $CH(x_{i_1}, ..., x_{i_{d+1}})$. For any $|\{i_1, ..., i_{d+1}\} = d+1$. For any x_i, define $H_i = \{v | v \cdot x_i \geq 1\}$. For any choice of $x_{i_1}, ..., x_{i_{d+1}}$, $0 \notin CH(x_{i_1}, ..., x_{i_{d+1}})$.
Separate by a hyperplane: $\exists u$ such that $u \cdot x > 1$ for all $x \in CH(x_{i_1}, ..., x_{i_{d+1}})$ in particular, $u \cdot x_{i_j} > 1$ for all $j \in [d+1]$.
So $0 \notin CH(x_{i_1}, ..., x_{i_{d+1}})$ implies $H_{i_1}, ..., H_{i_{d+1}}$ intersect (contains u).
So if $0 \notin CH(x_{i_1}, ..., x_{i_{d+1}})$ for any choice of $i_1, ..., i_{d+1}$, all choices of $d+1$, H_i's intersect. By Helly, all H_i's intersect. There exist $u, u \cdot x_i > 1$ for $i = 1, ..., M$. But this separates all x_i from 0 to a hyperplane, contradicting that $0 \in CH(x_1, ..., x_m)$. □

17 Matchings, sunflowers, and kernels (Mar 13)

17.1 Hall, Kőnig, and maximal matchings

Theorem 17.1. *Hall's Theroem: For a bipartite graph on (A, B)*
1. *$\exists$ a complete matching of A iff $\forall S \subseteq A, |\Gamma(S)| \geq |S|$ where $\Gamma(S) = \{v \in B | \exists u \in S, u \sim v\}$.*
2. *$\exists$ a matching with $|A| - k$ edges iff $\forall S \subseteq A, |\Gamma(S)| \geq |S| - k$*

Theorem 17.2. *Konig's Lemma: G bipartite implies $\mu = \tau$ where μ is the maximum matching size and τ is the minimum hitting set for G where a hitting set is a set of vertices that intersects all edges.*

Lemma 17.3. *For a general graph G, $\mu \leq \tau \leq 2\mu$.*

Proposition 17.4. *Any bipartite graph on m edges has a vertex of degree $\geq k$ or a matching of size $\geq k$ for $k = \lceil \sqrt{n} \rceil$*

Proof. Consider a maximal matching (μ edges). If $\mu \geq k$ we're otherwise $\tau = \mu < k$. Some vertex is min transversal intersecting $\frac{m}{\tau} > k$ edges. □

17.2 Sunflowers and the sunflower lemma

Definition 17.5. *H is a k-uniform hypergraph. A sunflower with r petals is a collection $f_1, ..., f_r \in H$ such that $\exists k$ such that each f_i is $f_i' \cup k$ for some f_i' with $f_i' \cap f_j' = \emptyset$ for $i \neq j$.*

Remark 17.6. *If $f_1, ..., f_r$ is sunflower, $\forall x \in V(H)$ either*
1. *$x \in f_i$ for all $i = 1, ..., r$ or*
2. *$x \in f_i$ for at most one i*

Definition 17.7. *$Sun(r, k) = \min m$ such that any k-uniform hyerpgraph on m edges has a sunflower on r petals.*

Lemma 17.8. *In a k-uniform hypergraph $\mu \leq \tau \leq k\mu$*

Proposition 17.9. *$Sun(r, k) \leq k!(r-1)^k + 1$*

Proof. Consider a matching of size μ in H which has m edges. I get a transversal of size $\tau = k\mu$. If $\mu \geq r$, we're done. Otherwise, I get a transversal of size $\tau = k\mu$. There is a vertex v such that v is in $\geq \frac{m}{\tau} \geq \frac{m}{k(r-1)}$ edges. Define a new hypergraph $H' = \{f \setminus v | f \in H, f \ni v\}$. If there is a r petal in H' then we have one in H as we can add v to the core. So

$$Sun(r, k) \leq k(r-1)Sun(r, k-1).$$

So the result follows by induction. □

Example 17.10. *k-Hitting Set Problem: Given H a k-uniform hypergraph and t, is there a hitting set of size $\leq t$.*
We define a hitting kernel for H as a hypergraph that has a hitting set of size $\leq t$ iff H does.
Claim: If H contains a sunflower $f_1, ..., f_r$ with $r = t + 2$, I can remove f_r from H.
I can make a kernel H' for H by removing all but $t + 1$ petals from every sunflower.
There are at most $m_{t,k} = k!t^k$ edges in any kernel for the hitting set problem with parameter t.

18 Uniform convergence for rectangles: symmetrization and growth (Mar 16)

18.1 Setup and symmetrization

Theorem 18.1. *Let $X = [0,1]^2$, and let F be a family of axis-parallel rectangles. Let $S_n = \{X_1, \ldots, X_n\}$ be a set of i.i.d. points randomly sampled from X. Define the true measure and the empirical measure respectively as:*

$$\mu(f) = \Pr[X_i \in f] \quad \textit{and} \quad \hat{\mu}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}_{X_i \in f}$$

We wish to bound the uniform deviation:

$$\Pr\left[\sup_{f \in F} |\hat{\mu}_n(f) - \mu(f)| > \epsilon\right]$$

Remark 18.2. *If we fix a specific point set S_n, there are at most n^4 different ways to partition the points using axis-parallel rectangles. However, this configuration depends entirely on the specific realization of the point set.*

Lemma 18.3 (Symmetrization)**.** *Assume $n \geq 2/\epsilon^2$. Let $S'_n = \{Y_1, \ldots, Y_n\}$ be an independent "ghost" sequence of i.i.d. samples, and let $\hat{\mu}'_n(f)$ be its empirical measure. Then:*

$$\Pr\left[\underbrace{\sup_{f \in F}(\mu(f) - \hat{\mu}_n(f)) > \epsilon}_{\textit{event } E}\right] \leq 2\Pr\left[\sup_{f \in F}(\hat{\mu}'_n(f) - \hat{\mu}_n(f)) > \frac{\epsilon}{2}\right]$$

Proof. Suppose event E occurs. We can choose some $f_E \in F$ witnessing this event, meaning $\mu(f_E) - \hat{\mu}_n(f_E) > \epsilon$. Because the ghost sample S'_n is independent of S_n, $\hat{\mu}'_n(f_E)$ has expected value $\mu(f_E)$ and variance bounded by $1/(4n)$.

By Chebyshev's inequality, the probability that the ghost sample's empirical measure deviates significantly from the true measure is:

$$\Pr\left[\mu(f_E) - \hat{\mu}'_n(f_E) > \frac{\epsilon}{2}\right] \leq \frac{1/(4n)}{(\epsilon/2)^2} = \frac{1}{\epsilon^2 n}$$

Since we assumed $n \geq 2/\epsilon^2$, this probability is $\leq 1/2$.

Fix any specific sequence $x_1, \ldots, x_n$ such that event E holds. For this fixed choice, $\mu(f_E) - \hat{\mu}_n(f_E) > \epsilon$. In particular, for these fixed events, if $\mu(f_E) - \hat{\mu}'_n(f_E) \leq \epsilon/2$, it must be that $\hat{\mu}'_n(f_E) - \hat{\mu}_n(f_E) > \epsilon/2$. Therefore:

$$\Pr\left[\hat{\mu}'_n(f_E) - \hat{\mu}_n(f_E) > \frac{\epsilon}{2}\right] \geq \frac{1}{2}$$

Integrating over all choices of $X_1, \ldots, X_n$ that lie in E, we get:

$$\Pr\left[\sup_{f \in F}(\hat{\mu}'_n(f) - \hat{\mu}_n(f)) > \frac{\epsilon}{2}\right] \geq \frac{1}{2}\Pr[E]$$

Multiplying by 2 yields the lemma. □

18.2 Proof: growth function and Hoeffding

Proof of Main Theorem. To prove the main result, we want to bound $\Pr\left(\sup_{f\in F}|\hat{\mu}_n(f)-\mu(f)|>\epsilon\right)$. We start with the one-sided version bounded by our symmetrization lemma:

$$\Pr\left(\sup_{f\in F}(\mu(f)-\hat{\mu}_n(f))>\epsilon\right)\leq 2\Pr\left[\sup_{f\in F}(\hat{\mu}'_n(f)-\hat{\mu}_n(f))>\frac{\epsilon}{2}\right]$$

Let Ω be the space of all choices of X_i's and Y_i's. We can generate a sample in Ω via two equivalent processes:

1. Choose $X_1,\ldots,X_n,Y_1,\ldots,Y_n$ i.i.d. uniform in $[0,1]^2$.
2. Choose $X_1,\ldots,X_n,Y_1,\ldots,Y_n$ i.i.d. uniform in $[0,1]^2$, then flip n random fair coins. If coin i is heads, swap X_i and Y_i.

We will use process 2 to generate Ω, relying only on the randomness of the coin flips. We will prove an upper bound that works for any fixed realization of pairs $\{X_1,Y_1\},\{X_2,Y_2\},\ldots$, utilizing only the random assignments (signs) of the coin flips.

Essentially, we are letting an adversary choose the $2n$ points, and we are using the randomness of the coin flips to "color" the points into the primary and ghost samples.

With respect to any fixed placement of these $2n$ points, there is a limited number of ways rectangles can classify them. Let $A=\{X_1,\ldots,X_n,Y_1,\ldots,Y_n\}$. Define the growth function $t_F(A)=\#\{f\cap A\mid f\in F\}$.

For the family F of axis-parallel rectangles, $t_F(A)\leq|A|^4$. Letting $t_F(n)=\max_{|A|=n}t_F(A)$, we only need this growth to be polynomial.

For any single configuration of points inside a fixed rectangle, applying Hoeffding's inequality over the random coin swaps bounds the probability of a deviation greater than $\epsilon/2$ by $2\exp(-n\epsilon^2/2)$. Applying a union bound over all $t_F(2n)$ effective rectangles gives:

$$\Pr\left[\sup_{f\in F}(\mu(f)-\hat{\mu}_n(f))>\epsilon\right]\leq 2t_F(2n)\cdot\exp\left(-\frac{n\epsilon^2}{2}\right)$$

To get the two-sided bound (the absolute value), we apply the exact same argument to the complement family $\overline{F}=\{X\setminus f\mid f\in F\}$, which shares the same growth function. Combining the two sides yields the final uniform convergence bound. □

19 VC dimension and the Sauer–Shelah lemma (Mar 18)

19.1 Shattering and VC dimension

Definition 19.1. *Let F be a family of subsets of a domain X. The* ***Vapnik-Chervonenkis (VC) dimension*** *of F, denoted d_F, is the size of the largest set shattered by F:*

$$d_F = \max\{d \mid \exists A \subseteq X, |A| = d, \text{ and } |F_A| = 2^{|A|}\}$$

where $F_A = \{f \cap A \mid f \in F\}$ is the restriction of F to A.

19.2 Sauer–Shelah Lemma

Theorem 19.2 (Sauer-Shelah)**.** *Let F be a family of subsets with finite VC dimension $d_F = d$. Let $t_F(n)$ denote the maximum number of subsets of any n-element set that can be formed by intersecting it with elements of F. Then:*

$$t_F(n) \leq \sum_{j=0}^{d} \binom{n}{j}$$

This implies a fundamental dichotomy for any concept class F: either $t_F(n) = 2^n$ for all n (if $d_F = \infty$), or there exists a finite d such that $t_F(n) = O(n^d)$.

Proof. We proceed by induction on n. Let $d_F = d < \infty$ be the VC dimension of F. We want to show that for any set A of size n, $t_F(A) \leq \sum_{j=0}^{d} \binom{|A|}{j}$.

Without loss of generality, let $A = [n] = \{1, 2, \ldots, n\}$. Let $F_{[n]} = \{f \cap [n] \mid f \in F\}$. By definition, $t_F(A) = |F_{[n]}|$.

To run the induction, we split $F_{[n]}$ by analyzing the elements with and without the element n. Define the projection onto the first $n-1$ elements:

$$F_0 = \{S \setminus \{n\} \mid S \in F_{[n]}\}$$

Note that F_0 is exactly $F_{[n-1]}$, the restriction of the family to the first $n-1$ elements. Next, define the set of "twins" where both the subset with n and without n are present in $F_{[n]}$:

$$F_1 = \{S \in F_{[n-1]} \mid S \in F_{[n]} \text{ and } S \cup \{n\} \in F_{[n]}\}$$

By construction, the size of our original family is the sum of these two sizes:

$$|F_{[n]}| = |F_0| + |F_1| = |F_{[n-1]}| + |F_1|$$

By the induction hypothesis, since the VC dimension of $F_{[n-1]}$ is at most d, we have:

$$|F_{[n-1]}| \leq \sum_{j=0}^{d} \binom{n-1}{j}$$

Now consider the VC dimension of F_1. If F_1 shatters a subset $B \subseteq [n-1]$, then by the definition of F_1, both the elements of F_1 and those same elements unioned with $\{n\}$ exist in $F_{[n]}$. Therefore, $F_{[n]}$ shatters $B \cup \{n\}$.

This implies that the VC dimension of F_1 must be strictly less than the VC dimension of $F_{[n]}$. Hence, $d_{F_1} \leq d-1$. Applying the induction hypothesis to F_1 yields:

$$|F_1| \leq \sum_{j=0}^{d-1} \binom{n-1}{j}$$

Adding these two bounds together, we get:

$$|F_{[n]}| \leq \sum_{j=0}^{d} \binom{n-1}{j} + \sum_{j=0}^{d-1} \binom{n-1}{j}$$

To combine these sums, we re-index the second sum to start at $j = 1$ (since $\binom{n-1}{-1} = 0$) and apply Pascal's identity, $\binom{n-1}{j} + \binom{n-1}{j-1} = \binom{n}{j}$:

$$\begin{aligned}
|F_{[n]}| &\leq \binom{n-1}{0} + \sum_{j=1}^{d} \left[\binom{n-1}{j} + \binom{n-1}{j-1} \right] \\
&= \binom{n}{0} + \sum_{j=1}^{d} \binom{n}{j} \\
&= \sum_{j=0}^{d} \binom{n}{j}
\end{aligned}$$

This completes the induction. □

20 Steiner Triple Systems (Mar 20)

Example 20.1. *Kirkman's Schoolgirl Problem: 15 students walk to school in groups of 3 and no two students ever in the same group twice. Can they go for 7 days?*

Definition 20.2. *A $STS(n)$ is a collection of triples $\mathcal{T}$ on $[n]$ such that $\forall x \neq y \in [n]$, $\exists! z$ such that $\{x, y, z\} \in \mathcal{T}$.*

Remark 20.3. *Any triple contains 3 pairs and any pairs is in 1 triple so $|\mathcal{T}| = (\binom{n}{2})/3 = \frac{n(n-1)}{6}$. Also, n must be odd because fixing x, $\{T \in \mathcal{T} | T \ni x\}$ induces a matching on $[n] \setminus \{x\}$. So $|[n] \setminus \{x\}|$ must be even.*

Lemma 20.4. *If $STS(n)$, then $n \equiv 1$ or $n \equiv 3 \pmod 6$.*

Remark 20.5. *For $n = 1$ there is none. For $n = 3$, trivial. For $n = 7$, the fano plane.*

Remark 20.6. *An $STS(n)$ is equivalent to a partition of $K - n$ into edge disjoint triangle. The vertex set of the triangles are an $STS(n)$.*

Theorem 20.7. *For even N, edges of K_n can be partitioned into $n - 1$ perfect matchings.*

Theorem 20.8. *If an $STS(n)$, then an $STS(2n + 1)$.*

Proof. Partition the points into two sets A and B of size n and $n + 1$ respectively. A is $STS(n)$ and B has n perfect matchings. For a vertex in A assign it matching in B. If a pair is A then take the triple in $STS(n)$. If they are both in B then take the matching and add the third point in A that is mapped to this matching. If they are different sets, take the matching in A and use it in B. □

Example 20.9. *An STS on $X = \mathbb{F}_2^n \setminus \{0\}$.*
Define triples $\{x, y, x + y\}$ $\forall x \neq y, x, y \neq 0$.
Fix any $x, y \neq 0$. Certainly, $x, y \in \{x, y, z\}$ for $z = x + y$. Suppose $x, y \in \{w, x, y\}$. WLOG suppose $y = w + x$ then $w = x + y$

Example 20.10. *An STS on $X = \mathbb{F}_3^n \setminus \{0\}$, $\forall x$ define a triple $\{x, x + d, x + 2d\}$. For any x, y, let $y = x + d$. We claim that $x, y \in \{x, x + d, x + 2d\}$.*